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Simulation of a Sustainable Heating System for an Industrial Greenhouse to Enhance the UK's Food Security

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Declaration of Originality and Submission Information

I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements.

Abstract

Background: Food security amid climate change and geopolitical uncertainty is a pressing societal and political challenge in the UK. Reducing food transport distance, carbon emissions, and logistic challenges, while maintaining conditions for the growth of 'warmer climate crops' are factors that the expansion of sustainable greenhouse farming within the UK could address.

Objectives: This study therefore investigates the feasibility of sustainable heating systems by examining and comparing three different heat pump(HP) sources for a theoretical UK-based industrial greenhouse growing a crop of avocados. Focusing on the comparison of air source(ASHP), shallow ground source (GSHP), and water source (WSHP) heat pumps, the study evaluates the different systems' energy efficiencies and power usage within the context of the operational heat demands for the crop's cultivation.

Methods: Utilising a self-constructed Python model for the simulation, six years of comprehensive weather data from Chelmsford, UK, are analysed and applied to heat transfer principles to simulate the thermodynamics of the system.

Results: The findings indicate that water sources are likely to be superior to air and ground sources of heat exchange in maintaining the optimal growth temperatures with lower energy consumption for this scenario.

Interpretation: The findings are further explored by calculating additional power requirements and how this would be sustainably supplied for the heat pump and additional requirements.

Conclusions: This research contributes to the field of sustainable agriculture by proposing a viable solution to improve domestic crop production in the UK using sustainable methods. A potential significant reduction in CO₂ emissions with improvement in food security, variety and availability could be achieved through widespread adoption of the proposed system. As with all modelling, the caveats include the need to progress through real-world testing and implementation to assess both the positive and negative wider impacts. Further modelling work could include an enhanced heat transfer model along with a Life cycle analysis (LCA) of the entire system to increase the reliability of the projected advantages of the proposed system.

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Table 1: Nomenclature

Definition	Abbreviation	Units
Air temperature	T_G	$^{\circ}\text{C}$
Ground temperature	T_G	$^{\circ}\text{C}$
Maximum external air temperature	T_{MA}	$^{\circ}\text{C}$
Minimum external air temperature	T_{MiA}	$^{\circ}\text{C}$
Maximum external ground temperature	T_{MG}	$^{\circ}\text{C}$
Minimum external ground temperature	T_{MiG}	$^{\circ}\text{C}$
Time value	t	hour
Heat transfer across wall/roof	Q_W	Watts (W)
Heat transfer across ground	Q_G	Watts (W)
Heat transfer from sun	Q_S	Watts (W)
Heat transfer total	Q_T	Watts (W)
Thermal conduction coefficient wall/roof	k_W	$\text{W/m}^{\circ}\text{C}$
Thermal conduction coefficient ground	k_G	$\text{W/m}^{\circ}\text{C}$
Thickness of wall/roof	d_W	m
Thickness of ground	d_G	m
Area of wall/roof	A_W	m^2
Area of ground	A_G	m^2
Current internal air temperature	T_i	$^{\circ}\text{C}$
New unregulated internal air temperature	T_{Ni}	$^{\circ}\text{C}$
Required heat addition	Q_c	Watts (W)
Desired internal air temperature	T_{di}	$^{\circ}\text{C}$
Regulated internal air temperature	T_{Ri}	$^{\circ}\text{C}$
Refrigerant enthalpy at current stage (i)	h_i	kJ/kg
Refrigerant entropy at current stage (i)	s_i	$\text{kJ/kg}^{\circ}\text{C}$
Refrigerant pressure at current stage (i)	P_i	kPa
Refrigerant temperature at current stage (i)	T_i	$^{\circ}\text{C}$
Mass flow rate above the flash tank	M_2	kg/s
Mass flow rate below flash tank	M_1	kg/s
Compressor 1 work input	W_{c1}	kW
Compressor 2 work input	W_{c2}	kW
Heat transfer between source and refrigerant at evaporator	Q_e	kW
Coefficient of performance	COP	N/A
Mass of internal air	m_a	kg
Specific heat capacity of internal air	$c_{p,a}$	$\text{J/kg}^{\circ}\text{C}$

1 Introduction

The first documented case of GH agriculture dates back to the Romans, when the need for controlled environments for crop cultivation became evident [1]. Initially, the idea was conceived to facilitate the growth of cucumbers as their emperor had fallen ill. Medicine at the time was built upon the fruits of nature and so he was prescribed a daily dose of cucumbers. However, the conditions of their surroundings were not suitable for their cultivation, Thus, a concrete room complete with a glass ceiling was constructed to replicate the Indian subcontinent's climate and give the emperor the best chance at recovery. GHs have since evolved into integral components of modern agriculture, enabling year-round production despite external conditions.

The acceleration of climate change and an unstable geopolitical landscape underscore the urgency for innovative solutions to enhance food security. This can be seen with high inflation since events such as Brexit and the Ukraine war. Furthermore, the impacts of climate change are already being seen with increased droughts and natural disasters in areas of previously high crop production. Numerous regions are threatened to become non-arable, leading to a global reduction in crop harvests. Projections by NASA indicate a 40% decrease in corn production within the tropics while the University of Edinburgh forecasts a staggering 50% decline across southern Europe under worst-case scenarios [2] [3]. These forecasts are alarming, as the repercussions would extend beyond the affected areas, impacting global food supply chains. According to the World Food Programme, 783,000,000 people are currently afflicted by chronic hunger [4], a figure that is expected to escalate with a projected 25% population growth by 2070 [5]. While these predictions are only considering how climate change will impact crop growth, the situation extends beyond this. The regions between the tropics are becoming increasingly inhospitable towards humans with many facing Sahara-esque conditions, this is threatening to displace over 3 billion people with many predicted to migrate to the global north. The combination of mass migration and decreased crop production leads to a worrying conclusion; We are running out of land and space that humans can occupy and cultivate.

The United Kingdom exemplifies these challenges by relying heavily on imports [6], as shown in Figure 1, despite ample resources for local production and being at low risk of climate change. The London School of Hygiene and Tropical Medicine has highlighted the UK's vulnerability to climate-induced disruptions in their food supply chain, showing that 33% of current imports come from moderately to highly vulnerable regions due to climate change [7].

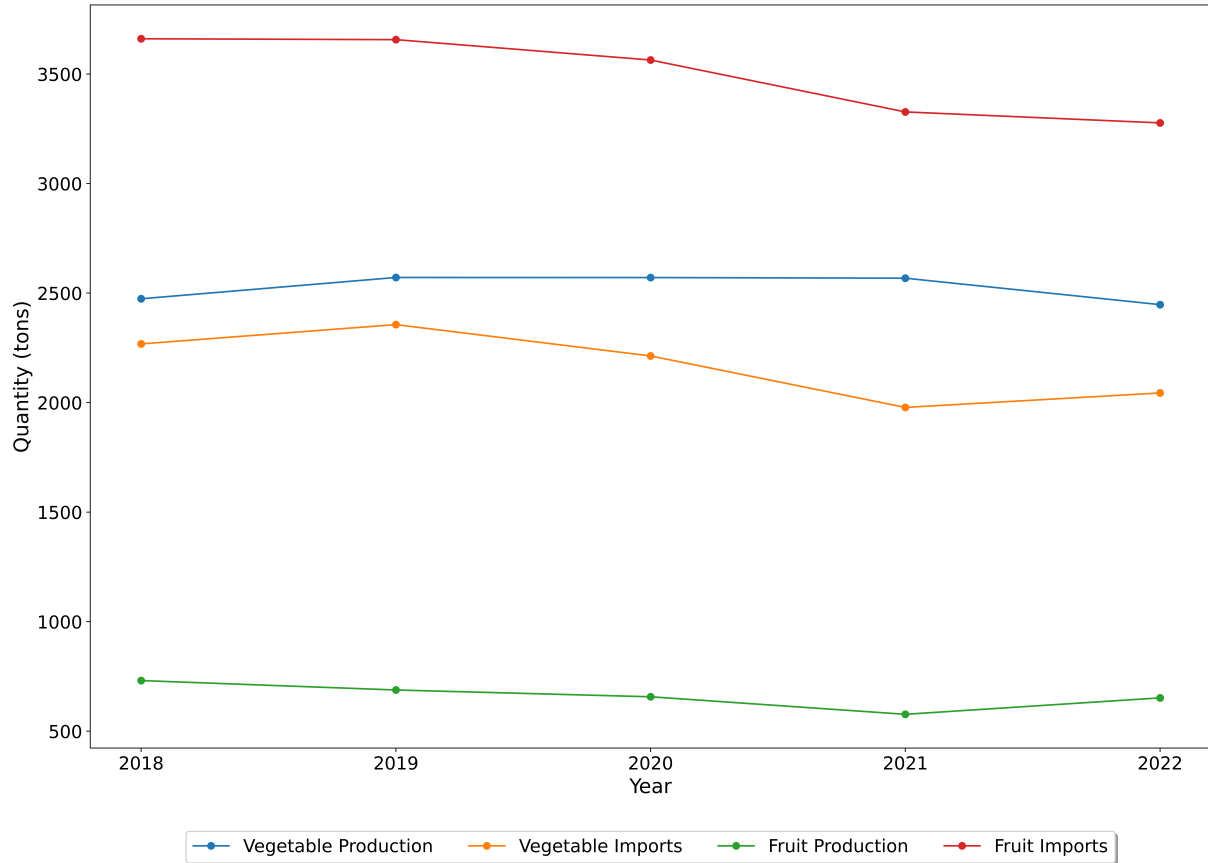


Figure 1: UK's Fruit and vegetable production vs imports

This project investigates the potential of industrial-scale GHs (GHs) in the UK, equipped with energy-efficient, sustainable technologies. This approach aims to harness sustainable heating methods to cultivate crops year-round, independent of external weather conditions. GH environments present a unique challenge compared to typical residential and commercial buildings because they are required to let sunlight in for photosynthesis, which constrains their use of standard insulation materials. Additionally, the harsh British winters limit the cultivation of tropical plants within traditional GH environments, restricting the growing season and the ability to self-produce the crops society demands. It is anticipated that heating will be the most significant energy consumer in industrial GH systems. To address this, heat pumps (HPs) have been selected for their energy efficiency and minimal environmental impact, aiming to provide a sustainable solution to meet heating demands.

This project aims to simulate an industrial GH with an integrated heat pump tailored to the UK's climate conditions. The project will carry out a simulation to determine the optimal HP source; air source (ASHP), shallow ground source(GSHP), and water source(WSHP), along with the power requirements and the performance of each HP. In doing so a comprehensive review of the feasibility of the system in the UK is hoped to be gained, along with the impacts on the food security of the UK. The simulation is conducted on a self-built Python model.

2 Background

2.1 Introduction

The foundations of this project are the laws of thermodynamics and heat transfer, along with the working mechanism of a heat pump. These will be outlined in detail before the main body of the report to provide a relevant description of the principles at play.

2.2 Laws and principles of thermodynamics

2.2.1 First Law of Thermodynamics

The first law of thermodynamics [8] states that energy cannot be created or destroyed, only transferred or converted from one form to another, this law is shown in Equation 1 and is known as the law of conservation.

$$\delta U = Q + W \quad (1)$$

2.2.2 Second Law of Thermodynamics

The second law of thermodynamics dictates the direction of heat transfer between two objects. Heat will naturally flow from objects or systems with higher temperatures to those with lower temperatures. This phenomenon is based on the principle of increasing entropy – as heat flows from hot to cold the total entropy of the system and its surroundings increases indefinitely. This is represented in Equation 2:

$$\delta S = \frac{\delta Q}{T} \quad (2)$$

2.2.3 Conduction and Convection

Conduction is the transfer of heat without any flow of matter [8], it is one of the fundamental mechanisms by which heat is transferred through solids, liquids and gases. It occurs when two solid bodies are in direct contact or within a single body with the existence of a temperature difference/gradient. Convection, on the other hand, occurs when a fluid (e.g. air or water) surrounds a solid body and transfers heat through bulk fluid motion. Newton's law of cooling described convection, stating that the rate of heat transfer is proportional to the temperature difference between the solid surface and the surrounding fluid, multiplied by the heat transfer coefficient h . This is represented in Equation 3:

$$Q = h * A * \delta T \quad (3)$$

2.2.4 Fourier's Law

Fourier's law of heat conduction is the primary foundation on which the simulation is built. Fourier's law quantifies the mechanism of conduction, stating that the rate of heat transfer through a material is directly proportional to the temperature difference across the material and its thermal conductivity. Essentially, heat flows more rapidly through materials with higher thermal conductivity and larger temperature differences [8]. Equation 4 below is the mathematical representation of Fourier's law of conduction:

$$Q = -k \frac{\delta T}{\delta x} \quad (4)$$

2.2.5 Latent Heat of Vapourisation and Condensation

During phase changes refrigerants absorb or release latent heat without changing temperature, this process is governed by the principle of latent heat, which describes the amount of heat absorbed or released during a phase change at a constant temperature.

2.3 Heat Pump Working Mechanism

Put simply HPs transfer energy from a heat source (HSo) to a heat sink (HSi) with the use of a medium, namely a refrigerant. During this process the refrigerant undergoes phase changes, compression and expansion, to amplify the absorbed heat energy and release a larger amount.

2.3.1 Refrigerant

The role of the refrigerant in a heat pump system is integral to the overall purpose. As it circulates through the system the refrigerant undergoes phase changes and temperature variations to facilitate efficient heat transfer, while adhering to the first law of thermodynamics 2.2.1. Refrigerants are selected based on their thermodynamic properties, successful refrigerants must have a high latent heat of condensation and vapourisation. This is so that the refrigerant absorbs a high amount of heat when it changes from a liquid to a vapour and releases the majority of its heat at the condenser where it will reject the heat to the target entity.

2.3.2 Vapor Compression Cycle

The vapour compression cycle is the fundamental thermodynamic process that occurs during the operation of a heat pump [9]. It consists of four stages: compression, condensation, expansion and evaporation.

Compression:

In this stage, the low-pressure vapour refrigerant from the evaporator enters the compressor. The compressor then increases the pressure and temperature of the refrigerant through compression turning it into a high-pressure, high-temperature vapour. This stage requires input energy, typically in the form of mechanical work, to increase the enthalpy of the refrigerant.

Condensation:

After leaving the compressor, the high-pressure, high-temperature vapour refrigerant flows into the condenser. In the condenser, the refrigerant releases heat to the surroundings or to another medium, causing it to condense into a high-pressure liquid. As the refrigerant condenses, its temperature and pressure decrease while transferring heat to the surroundings.

Expansion:

The high-pressure liquid refrigerant exits the condenser and enters the expansion valve. In this stage, the refrigerant undergoes a rapid expansion, causing a significant drop in pressure. This sudden decrease in pressure results in a decrease in temperature, leading to the formation of a low-pressure, low-temperature mixture of liquid and vapour.

Evaporation:

The low-pressure, low-temperature mixture of liquid and vapour refrigerant enters the evaporator. Here, the refrigerant absorbs heat from the surroundings, such as air or water, causing it to evaporate into a low-pressure vapour. This vapour is then drawn back into the compressor to begin the cycle anew.

The vapour compression cycle is a simple overview of a HP system. However, in many scenarios, it becomes advantageous to create a multistage-stage HP system, these often increase the overall performance of the HP.

3 Literature Review

3.1 Introduction

This literature review aims to analyze the following:

1. The current state of the agricultural industry
2. The expansion of GH technologies
3. Existing case studies on heat pumps

By examining these areas, a comprehensive understanding of the context for investigating the feasibility of utilizing a heat pump in controlled environment agriculture (CEA) with the UK's climactic conditions can be explored.

3.2 Agricultural Impacts on Climate Change

The global agricultural industry significantly contributes to environmental degradation, with emissions from land use changes and machinery operations playing a substantial role, Pendrill et al. estimated that agriculture emissions account for 15% of man-made CO₂ emissions[10]. A study on deforestation by the agriculture and forestry industry by Pedrill et al. analysed the impacts that our increasing land use for agriculture and forestry had on the world's climate [10]. Pedrill et al. found in the tropics alone that the expansion of agriculture and forestry contributed to net emissions of 2.6 Gigatonnes of CO₂ per year, with palm oil and soybeans contributing 0.6 Gigatonnes of CO₂ per year. Furthermore, the study found that roughly 6% of embodied emissions from deforestation in the tropics were traded from Papua New Guinea to the UK in the form of vegetables fruits and nuts. The UK as a nation has already gone through its deforestation and only 13% remains [11], however, we are not making the most of this open space for agriculture efficiently, and still importing a vast majority of our fresh produce from countries that remain densely forested and undergo deforestation to provide our goods.

3.3 Avocado Example

Blaauw et al. conducted a study that tracked the life cycle of an avocado from the fields in South Africa to our tables in the UK [12]. The study considered the implications at every stage from planting through ripening, packhouses and shipping. They collate the different types of fuel used along the way before applying a conversion factor to globalise the results, their common unit for each stage is $kg/CO_2e/tonne$. The first stage analysed is farming, from seed preparation to harvesting, which encompasses fertilizers, pesticides and machinery. Blaaw et al. found that the total emissions per tonne of avocados shipped from South Africa to Nottingham were 904.31 kg/CO_2e . This Lifecycle analysis serves as a benchmark for comparing the unsustainable farming practices and shipping emissions of a real-world scenario with the results of this simulation.

3.4 Greenhouse Usage Globally

The escalating demand for fresh out-of-season agricultural produce has spurred the expansion of GH production on a global scale. CEA techniques, which involve precise control over environmental variables like temperature, humidity, and light exposure, have been shown to significantly enhance crop yields while optimizing resource utilization year-round [13]. A study by

Talbot et al. models the growth rates of multiple crops within CEA under different conditions, providing evidence of its pivotal role in efficient crop production[14]. A study by Azaza et al., underscores the efficacy of a GH Internet of Things (IOT) monitoring system in maintaining specified temperature and humidity levels, resulting in a noteworthy reduction of energy consumption by approximately 25% and water consumption by 33% [15]. Despite the recognized benefits of CEA, research often limits its scope to well-known crops like lettuce, leaving gaps in understanding of the broader applicability and optimization of CEA systems.

In regions like Iran, where GH cultivation thrives, dependence on diesel fuel for temperature control poses environmental and economic challenges. Comparative studies reveal discrepancies in the cost-effectiveness of heating systems like natural gas boilers versus heat pumps, influenced by varying energy costs across regions[16]. While Iran may face economic barriers to transitioning to more sustainable heating methods, regions like the Netherlands demonstrate the potential cost benefits of adopting heat pump technology.

3.5 Heat Pump Case Studies

HPs offer a promising alternative to traditional fossil fuel heating systems, presenting a more sustainable and environmentally friendly option for meeting heating demands. Numerous studies have compared the environmental and economic performance of HPs to fossil fuel heating systems, revealing compelling advantages in favour of HP technologies.

A study comparing ASHP and GSHP, conducted a life cycle analysis (LCA) to evaluate their environmental impacts over time. The GSHP emerged as the more environmentally favourable option, demonstrating lower negative environmental impacts compared to its ASHP counterpart [17].

In another study, the integration of photovoltaic (PV) solar cells with GSHP showcased significant advantages over liquefied natural gas (LPG) heating systems in terms of both environmental impact and cost-effectiveness [18]. This integrated system outperformed LPG heating systems, emphasizing the potential of renewable energy technologies in conjunction with heat pumps to achieve sustainable heating solutions.

Furthermore, research focusing on reducing the environmental impact of heating in the European Union (EU) [19] found that ASHP significantly outperformed natural gas boilers in terms of GH gas (GHG) emissions. Building energy consumption, a major contributor to GHG emissions was substantially reduced by up to 54% with the adoption of ASHP, highlighting their role in mitigating climate change impacts.

Examining the performance of GSHP and ASHP in extreme climates, studies conducted in Russia [20] and Turkey [21] provided valuable insights into their respective efficiencies. In the coldest Russian cities, GSHP exhibited higher coefficients of performance (COP), making them more efficient options for heating in extremely low temperatures. Similarly, in Turkey, GSHP were found to be 11.09% more economical than ASHP, underscoring their cost-effectiveness in specific geographic contexts. The comparative analysis of heat pump technologies demonstrates their superiority over fossil fuel heating systems in terms of environmental impact, cost-effectiveness, and efficiency. These findings underscore the potential of heat pumps as a sustainable solution for meeting heating needs while reducing reliance on finite fossil fuel resources and mitigating climate change effects.

3.6 Refrigerant

The study conducted by Wang et al. investigates the environmental performance of a GSHP system using 5 different refrigerants of low global warming potential (GWP), across different climate zones [22]. Throughout the study, the environmental impacts of operating the GSHP along with the impacts of the refrigerants are combined to provide a comprehensive assessment of the system as a whole. Wang et al. found that refrigerants R152a (GWP 124) and R134ze(E) (GWP 7) are the optimal low-GWP refrigerants across the various climate zones, While R290 (GWP 3) and R1234yf (GWP 4) are not as effective due to the systems total emissions when employing these refrigerants. While the study conducted by Wang et al. is beneficial, the use of ammonia as a refrigerant offers promising findings. Soujoudi et al. analysed the use of ammonia as a part mix in larger industrial pre-cooling systems [23]. It was found that the Coefficient of performance (COP) was enhanced with the addition of ammonia while reducing operational costs and specific energy consumption. Furthermore, a study conducted by Lima et al. analyses the role of ammonia as a sustainable refrigerant [24]. In a solution combined with lithium nitrate, this combination results in a soluble efficient refrigerant mix, minimizing corrosion of the refrigeration system. Lima et al, emphasize the ability of the ammonia-based refrigerant to operate at lower temperatures than other refrigerants, a key property when analysing HP systems across varying cold climates. The International Institute of Ammonia Refrigeration (IIAR) found that the thermodynamic properties of ammonia are 3-10% more efficient than that of other commercial refrigerants [25]. While there are negatives in the use of ammonia due to the hazards it poses to humans, however, the IIAR created a code for the safe and proper handling of ammonia in refrigeration to minimise these risks.

3.7 Conclusion

The traditional agricultural industry has devastating impacts on the environment with deforestation and outdated technologies. Modern CEA technology promises to be an innovative method in reducing the environmental impacts of agriculture while increasing crop yields, however, studies into their effectiveness in a wide array of climate conditions are limited. There is also an absence of investigations into different heating mechanisms employed by CEA to reduce their environmental impact. The case studies reviewed in this section provide substantial evidence HPs can prove economic even in the coldest environments. Questions remain, however, as to their capabilities in maintaining optimal conditions for CEA in cooler climates such as the UK.

4 Heat Pump Design

This section will discuss the selected heat pump model and the reasoning behind it. As discussed in section 2.3.2 The vapour compression cycle is a simplistic overview of a HP. For the large-scale heating that is anticipated in this simulation, a more complex model has been employed. It is essentially a multistage VCC. The schematic diagram for the chosen heat pump is shown below in Figure 2.

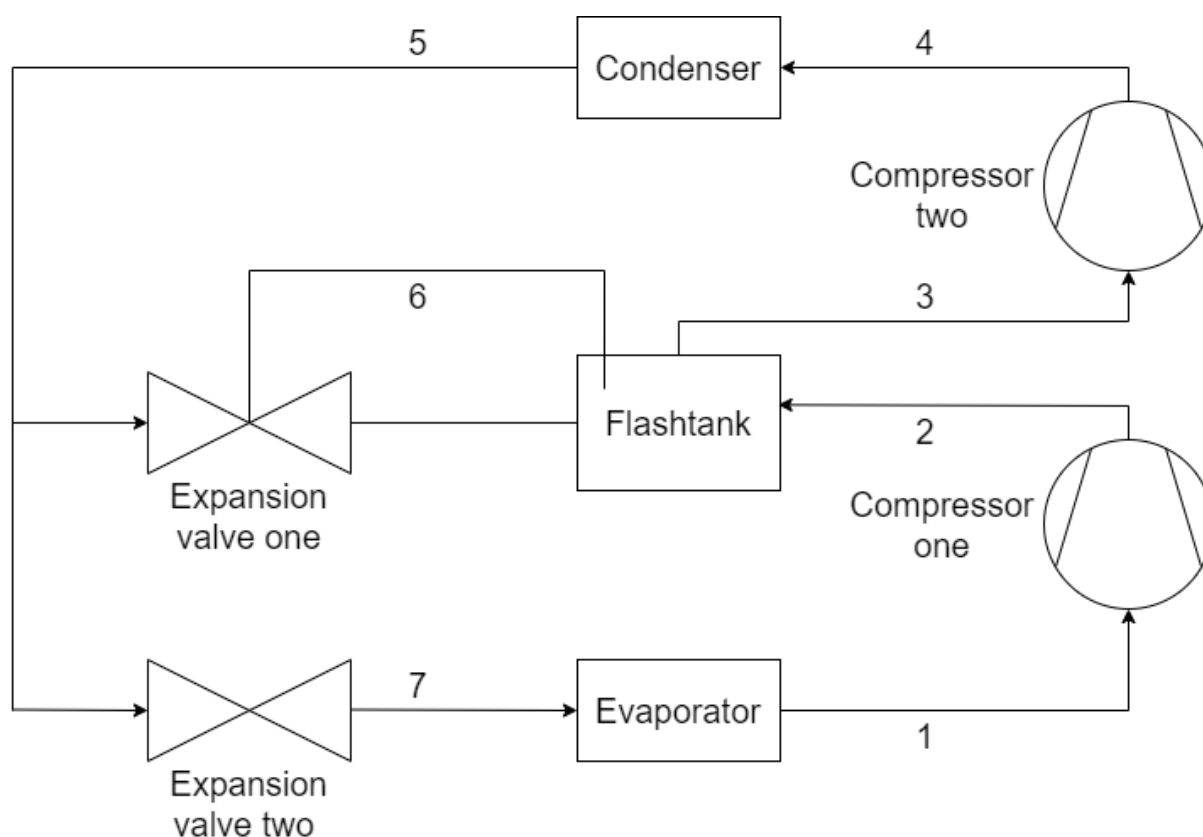


Figure 2: HP Schematic Diagram

A dual VCC has been employed to increase the overall efficiency of the system. The drawback of single-stage VCC is their fixed capacity. Unlike single-stage, multistage HPs can adjust their output to match the required heating load. This results in the ability to ramp up or slow down heating during the fluctuating temperature of the UK. Ultimately leading to a reduction in energy consumption along with more precise control of the indoor temperature. This is essential for the efficient use of CEA.

4.1 Source Temperatures

The source temperatures for the ASHP are the external air temperatures.

The source temperatures for the GSHP are taken from 4 inches below the ground, this is different to a borehole GSHP as this is cheaper to install but may have longer energy consumption costs. it is known as a shallow GSHP.

The source temperatures for the WSHP are the river Severn water temperatures.

4.2 Heat Pump System Process

The multistage HP system with a flash tank intercooler possesses several crucial components working together to ensure efficient heat transfer and energy conversion. At the system's heart lies the evaporator, where the refrigerant absorbs heat from the HSo, typically the ambient air or a thermal reservoir, and undergoes a phase change, transitioning into a vapour. This vapour, with the absorbed heat energy, moves on to the first compressor. Here, the vapour is compressed, leading to an increase in temperature and pressure, resulting in superheating.

The above schematic (Figure 2) employs a flash tank intercooler between compression stages. This acts as a heat exchanger, operating by cooling the superheated vapour from the first stage of compression using a medium such as air or water. In doing so this causes a drop in temperature and volume while increasing the fluid density before the second stage of compression. This leads to a higher mass of vapour entering the compressor without exceeding the compressor's volumetric capacity, making it more suitable for demanding heating loads.

After leaving the flash tank the vapour enters the second compressor, where it undergoes further compression to elevate its temperature to the requirements of the condenser. Once again, the refrigerant becomes superheated before entering the condenser.

At the condenser, the refrigerant undergoes a heat exchange process with the radiator water, which serves as the HSi for heating the internal air. As heat is transferred from the refrigerant to the water, the refrigerant's temperature decreases, causing it to condense into a liquid state. This high-pressure liquid then cycles past two expansion valves to the evaporator to begin again. At the first expansion valve, a portion of the vapour remaining in the mixture is redirected through an expansion valve back to the flash tank to maintain system stability and efficiency. Meanwhile, the remainder of the refrigerant, possessing lower entropy and pressure, passes through a separate flash valve to re-enter the evaporator as a liquid. Through this intricate process, the heat pump system efficiently transfers heat energy from the HSo to the HSi, providing effective heating while maintaining energy efficiency.

4.3 Refrigerant Selection

The chosen refrigerant for the heat pump system is ammonia. The key objective of this report is to maintain sustainability. Ammonia as a refrigerant has an ozone depletion potential of 0 and a GWP of 0, while these properties are excellent, attention must be paid to ensuring there are no leaks in the system as ammonia present in nature (soil and water) can be harmful to the environment and humans.

The properties of ammonia are well suited to being a refrigerant, previously discussed in Subsection 3.6, as it possesses high latent heat of vapourisation, Section 2.2.5, and a low boiling point. This means it can absorb and release large amounts of heat efficiently. The properties of ammonia compared to other refrigerants from Section 3.6 are shown below.

Refrigerant	LHoV (kJ/kg)	BP (°C)	ODP	GWP
Ammonia	1371.2	-33.5	0	0
R32	382	-52	0	675
R134a	92.8	-26.1	0	1300

Table 2: Ammonia properties

The evidence in Table 2 that ammonia is the most well-suited to the sustainability theme

is obvious here with a GWP of 0 compared to the others while also possessing a much higher latent heat of vapourisation.

4.4 Heat Pump Assumptions

1. Ammonia is saturated at the evaporator, flash tank, and condenser
2. Losses at the condenser are negligible
3. Compression is isentropic
4. Heat pump components working at 100% efficiency

The assumptions made relate closely to the foundational laws of thermodynamics and principles of heat transfer, as discussed in Section 2. The reasoning below corresponds to the numbered assumptions above.

1. Assuming saturated ammonia at critical points simplifies the extraction of its thermodynamic properties to aid in accurate phase change calculations. This assumption applies the first law of thermodynamics, Section 2.2.1, focusing on energy conservation during these calculations and aids in straightforward assessments under the second law, Section 2.2.2, where predictable changes occur during vapourisation and condensation.
2. By assuming negligible losses at the condenser we simplify the energy balance, accentuating the effective use of extracted heat. This assumption supports the First law of thermodynamics by ensuring all extracted heat is accounted for in the heating process or environmental discharge, providing an idealised efficiency metric for the system.
3. The isentropic compression assumption treats the operation of the compressor as adiabatic and reversible, simplifying the energy and work calculations by ignoring heat exchange with the environment and entropy production. This aligns with the first law of thermodynamics in calculating work based on internal energy changes and supports the second law by assuming an ideal scenario where entropy remains constant, allowing the extraction of thermodynamic properties in the superheated state.
4. The components working at 100% efficiency assumption must be made as the choice of the components

By applying these assumptions the simulation aligns with the laws of thermodynamics and heat transfer, to allow for an accurate theoretical understanding of the efficiency and performance of the HP system, providing an idealised baseline from which real-world applications can be developed.

4.5 Refrigerant Property Extraction

This section will detail how the properties of ammonia were extracted from the relevant, superheated and saturated tables [26] based on the assumptions and applications of the principles of thermodynamics detailed in section 2. Properties are extracted based on their corresponding row to the known values at each stage, and the refrigerant state, whether saturated liquid/vapour, or superheated vapour. This extraction process is shown below in Table 3

Stage	State	Temp (°C)	Pressure (kPa)	Enthalpy(kJ/kg)	Entropy
1	Saturated vapour	T_{source}	Extracted	Extracted	Extracted
2	Superheated vapour	Extracted	$=P_3$	Extracted	$=s_1$
3	Saturated vapour	Extracted	$=\sqrt{P_1 * P_5}$	Extracted	Extracted
4	Superheated vapour	Extracted	$=P_5$	Extracted	$=s_3$
5	Saturated liquid	80	Extracted	Extracted	Extracted
6	Saturated liquid	$=T_1$	$=P_3$	$=h_5$	$=s_{3,liquid}$
7	Saturated liquid	$=T_1$	$=P_1$	$=h_5$	$=s_{1,liquid}$

Table 3: Refrigerant properties extraction

Below is a graphical representation of the properties of ammonia at each stage of the heat pump, shown in Figure 2, with the applied assumptions and extraction techniques described in Table 3.

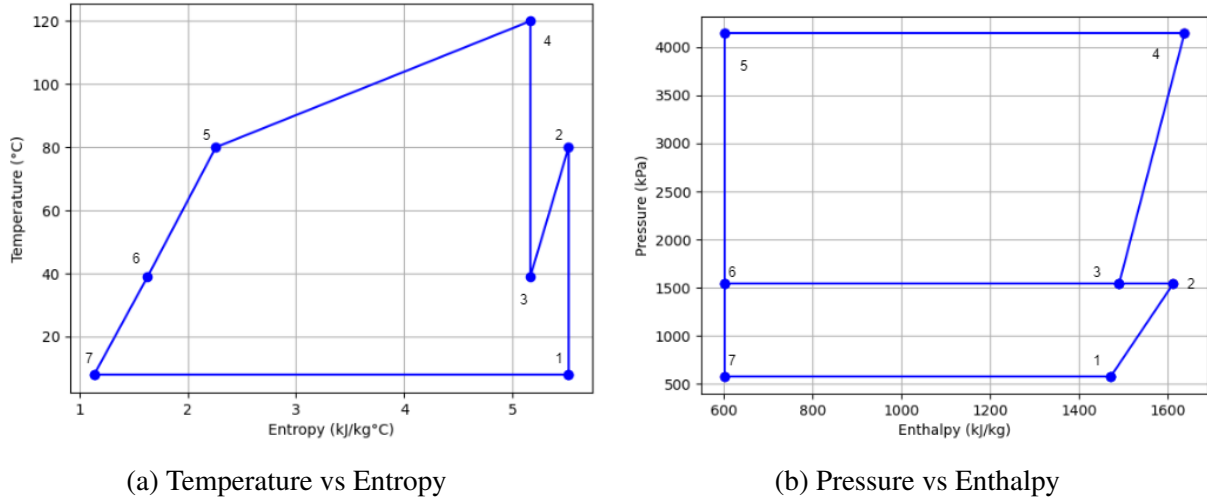


Figure 3: Refrigerant properties over stages of the heat pump.

4.6 Heat Pump Parameter Calculations

The following section, using the thermodynamic principles discussed in Section 2, will display the relationships between the refrigerant properties and the performance parameters of the HP displayed in Figure 2.

$$M_2 = \frac{Q_c * 0.001}{h_4 - h_5} \quad (5)$$

$$M_1 = \frac{M_2 * (h_3 - h_5)}{h_2 - h_5} \quad (6)$$

$$W_{c1} = M_1 * (h_2 - h_1) \quad (7)$$

$$W_{c2} = M_2 * (h_4 - h_3) \quad (8)$$

$$Q_e = M_1 * (h_1 - h_7) \quad (9)$$

$$COP = \frac{Q_c * 0.001}{W_{c1} + W_{c2}} \quad (10)$$

In Equation 5 & Equation 10 the Q_c value is multiplied by a value of 0.001 as in the simulation it is originally calculated as Watts, when the HP parameters are all dealt with in kiloWatts. The values used in these calculations can be found in appendix A.

5 Simulation and Method

5.1 Simulation

The following section details how the information gathered in the background and literature review has been employed to analyse the feasibility of a CEA while comparing ASHP, GSHP and WSHP as the heating unit for temperature control in the UK's climate.

5.1.1 Simulation Overview

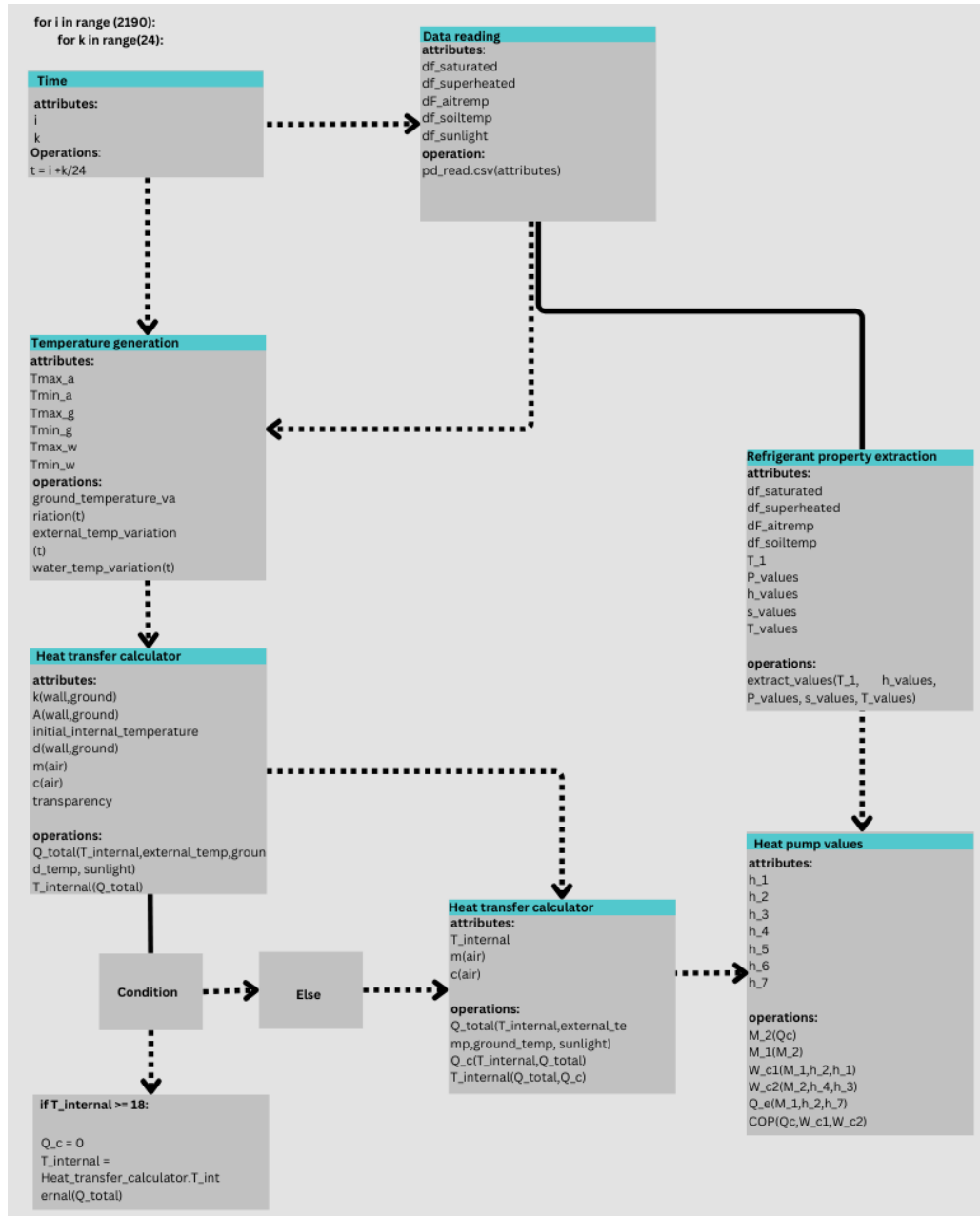


Figure 4: Unified modelling language (UML) diagram

The UML diagram provided in Figure 4 illustrates an overview of the simulation process, where solid lines without arrows denote an association, and dashed arrows represent an implementation. Association means the classes involved communicate with each other, or that one class

uses/interacts with the services of the other. Implementation denotes one class realizing or carrying out the behaviour specified by another class. Before the simulation's commencement the libraries 'numpy', 'pandas' and 'matplotlib' were imported, and the required data including refrigerant properties and weather information were loaded from CSV files. These files contained daily ground and air temperatures, and the intensity of sunlight. The refrigerant properties CSV file contained all necessary values for superheated and saturated ammonia.

The simulation begins with the initialisation of a for loop, to represent the hourly time intervals. Subsequently, the time is implemented into the 'Data reading' class where the current day's high and low temperatures are extracted. These are then implemented into the 'Temperature generation' class alongside the time interval, to determine the temperature values of soil, air and water. The resulting air and ground temperatures are implemented into the 'Heat transfer calculator' class to compute the heat losses and thus update the internal temperature. This internal temperature is then checked against a condition; if met the heat pump remains inactive, maintaining the current internal temperature. Conversely, if the condition is not met the internal temperature is recalculated within the 'heat transfer calculator' class incorporating the necessary heat addition to maintain the desired internal temperature. The calculated heat addition is then implemented into the 'heat pump values' class to assess the heat pump's performance during the specified time interval. This evaluation is completed by implementing the values obtained in the 'refrigerant properties extraction' class. This completes the iterative simulation process.

5.2 Method

5.2.1 Site Selection

In selecting the site for the GH, several factors were taken into consideration. The primary concern was the availability of accurate weather data spanning a minimum of 5 years. This extended timeframe was chosen to provide a robust data set for determining the feasibility of the project over a prolonged period. Additionally, The site selected required rail freight connections to the rest of the country, facilitating efficient mass transportation and thereby reducing the overall carbon footprint. Moreover, it was essential for the chosen location to have access to a river that would act as a viable source for the heat pump system. Table 4 presents a list of the potential sites that were shortlisted, along with the criteria and their respective performances in said areas.

Potential site	Weather data	Freight network	River
Merseyside	No Sunlight data	2-2.5 million tonnes	Mersey
Chelmsford	Yes	>4 million tonnes	Crouch
Gloucester	Not accessible historical record	0.5-1 million tonnes	Severn
Scunthorpe	Not accessible data sets	>4 million tonnes	Trent

Table 4: Site selection

The sites in the table were gathered upon research. Among the listed sites, Chelmsford is the most suitable location for the simulation, possessing historical weather data that encompasses sunlight (kWh) and daily air and ground temperatures. This is not to say the other locations are not feasible for such a system in the real world, but only that the historical weather data for the required simulation is not present. Moreover, Chelmsford benefits from a Freight rail network

with a high capacity and short distance to London enabling efficient redistribution to the rest of the country as seen in Figure 5 [27].

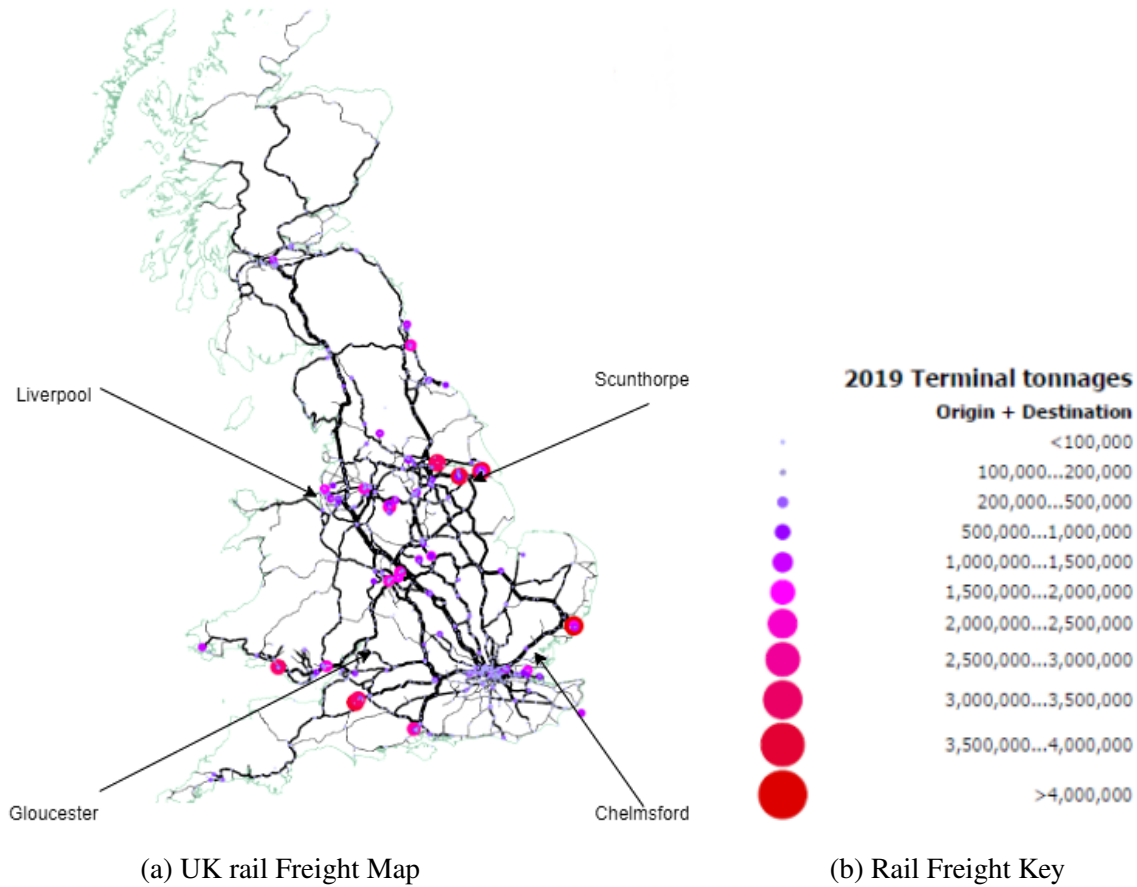
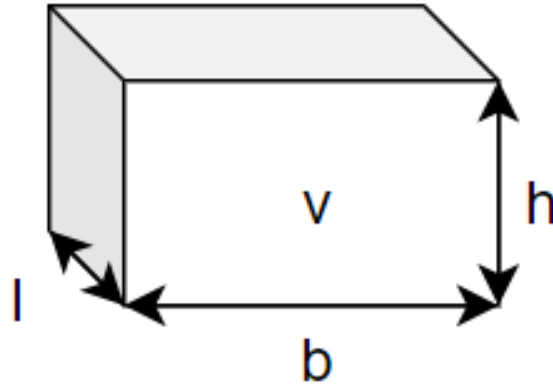


Figure 5: Uk Rail Freight

5.2.2 Greenhouse Parameters

The next phase in constructing the model involves defining the GH parameters. To establish an accurate heat transfer model complete knowledge of the system's properties is essential. Within the agricultural industry, the largest GH structures typically span up to $400m^2$ in area [28]. Consequently, a $20 \times 20m$ floor plan was deemed suitable for this analysis. Moreover, the specified height of $5m$ accommodates the growth requirements of taller plants, such as avocados. It is noteworthy that exceeding this size would make the challenge of maintaining optimal temperature conditions increasingly difficult. Hence, if this were to be implemented instead of opting for a single larger GH, a modular approach involving multiple sub-units within a larger area would be used. The dimensional representation of the GH is depicted below in Figure 6a along with a table of the assigned values in Figure 5b.



(a) Dimensional representation of the Greenhouse

base (b)	length (l)	Height (h)	Volume (v)
20m	20m	5m	$2000m^3$

(b) Greenhouse dimensions

Figure 6: Greenhouse model

5.2.3 Materials

With the dimensions of the GH established, the next step entails selecting appropriate materials for constructing the model. This consists of identifying materials that offer desirable properties such as thermal conduction coefficient and transparency while ensuring that the size of panels is commercially viable. Initially, the simulation was run with $15cm$ thick concrete slabs for flooring and $5mm$ thick polycarbonate (PCB) panels for the walls and roof. The relevant properties, obtained from the engineering toolbox [29], are shown in Table 5

Material	Assignment	k ($W/m^{\circ}C$)	Transparency (%)	Thickness (d, m)
Polycarbonate	Wall/Roof	0.19	0.9	0.01
Concrete	Floor	0.7	N/A	0.15

Table 5: Material properties

These materials are commonly employed in basic GH systems and are therefore commercially available, however when the simulation was conducted the heat losses, and therefore the heating requirements were too large. Therefore the following materials were selected to improve the thermal insulation of the system. PVC has a wide range of thermal conductivities found in papers and data sheets [30], [29], [31]. A value of $0.12W/m^{\circ}C$ was selected from the range as a median, furthermore, natural rolled rubber is used commonly in GHs with its properties taken obtained via [29], [32]. These materials show a decreased k value and therefore increased insulation leading to a more energy-efficient system. Table 6 shows the values for the new materials employed in the system.

Material	Assignment	k (W/m°C)	Transparency (%)	Thickness (d, m)
Polyvinyl chloride	Wall/Roof	0.12	0.9	0.01
Natural rolled rubber	Floor	0.17	N/A	0.15

Table 6: Material properties

5.2.4 External Conditions Generation

With the site and GH model established, the next step involves generating the external conditions using the weather data obtained from Chelmsford weather station, this serves as the foundation for the heat transfer model. Realistic weather data is imperative for accurately assessing the feasibility of the GH system. The temperature data obtained from the Chelmsford weather station provides only the maximum and minimum recorded temperatures for each day [33]. To emulate real-world conditions these values were input into a sine function replicating a typical day-night cycle.

$$T_A = \left(\frac{T_{MA} + T_{MiA}}{2} + \frac{T_{MA} - T_{MiA}}{2} \cdot \sin \left(\frac{2\pi t}{24} - \frac{\pi}{2} \right) \right) \quad (11)$$

$$T_G = \frac{(T_{MG} + T_{MiG})}{2} + \frac{(T_{MG} - T_{MiG})}{2} \cdot \sin \left(\frac{2\pi t}{24} - \frac{\pi}{2} \right) \quad (12)$$

After extracting and manipulating the daily highs and lows for both ground and air using Equations 11 & 12, the result is depicted in Figure 7.

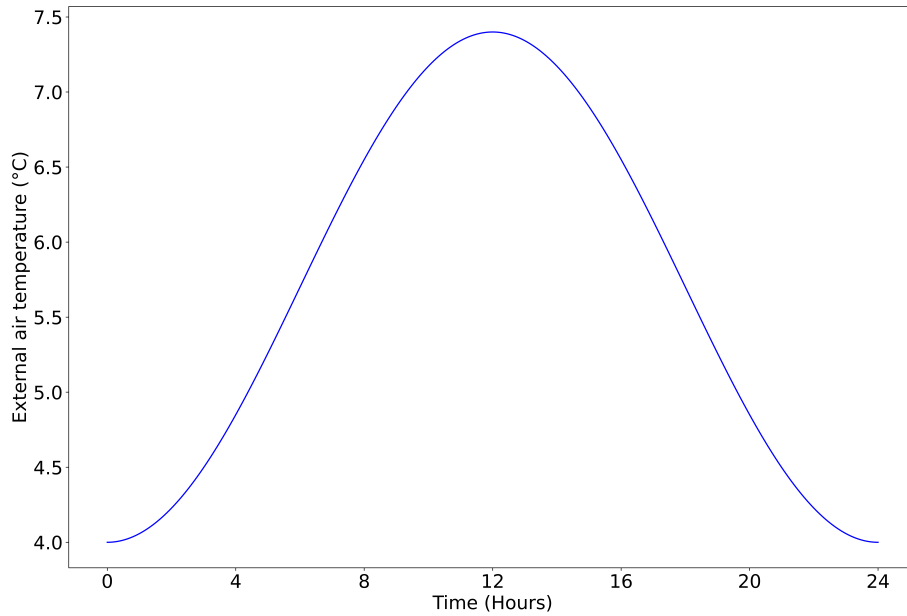
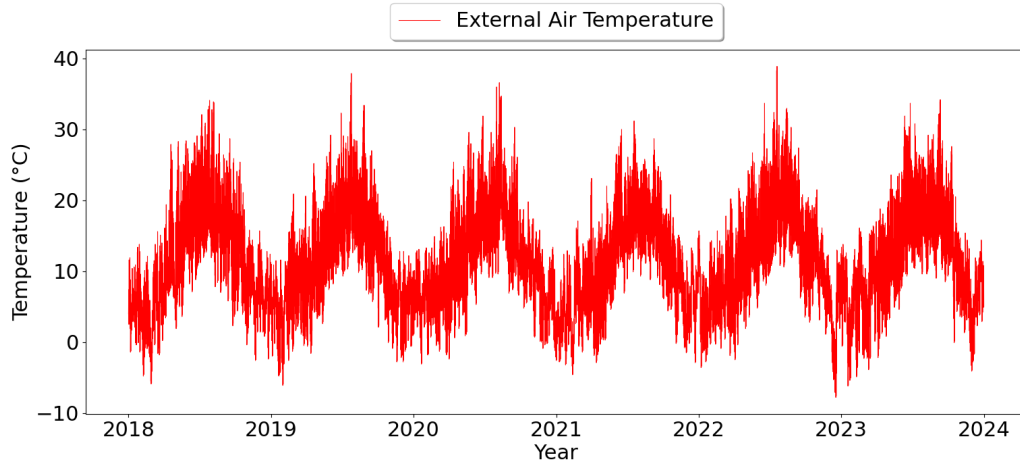
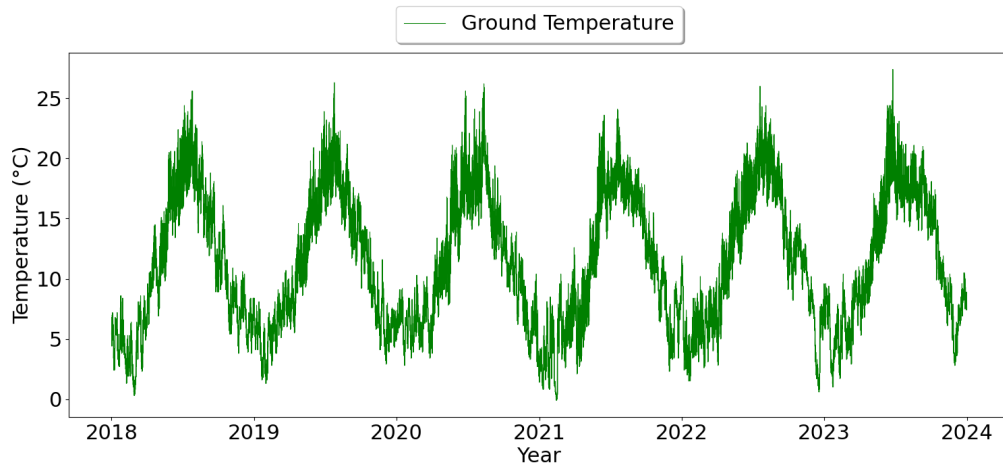


Figure 7: Air temperature variation over 1 day representing Equation 11

This process was extended over 6 years, With each iteration occurring hourly. The daily highs and lows were extracted from a CSV file using pandas, and then input into Equation 11 & Equation 12. The outcome is exhibited below in Figure 8.



(a) External Air Temperature Profile



(b) External Ground Temperature Profile

Figure 8: External Temperatures

In Figure 8, the graph illustrates distinct seasonal variations each year, signifying an accurate temperature generation model. Furthermore, the graph highlights the comparatively stable nature of ground temperature, Figure 8b, exhibiting fewer erratic daily and seasonal fluctuations.

The temperature data for rivers across the UK is limited. However, water's physical properties mean variation throughout the year is less so than that of the air and ground. The maximum water temperature is felt in September and the minimum is in March. With this data collected from [34] a sine wave manipulation was applied to create water temperatures that correspond to the timing of the highs and lows of the river Crouch, this is presented in Figure 9.

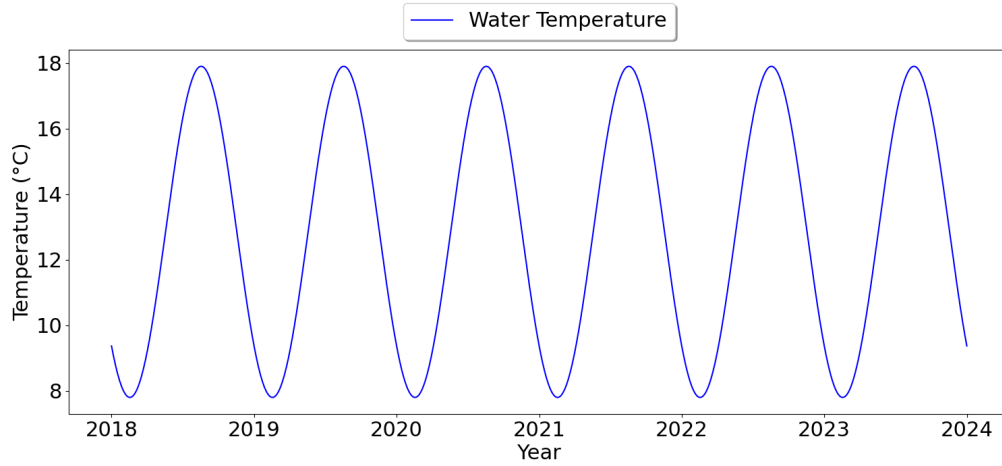


Figure 9: River Crouch Water Temperature

5.2.5 Assumptions

Before calculations surrounding the system can be conducted, assumptions must be made. The reason is that without a practical experiment that would be costly without guaranteed success, a hyperrealistic simulation was not feasible. Therefore to simplify the system while maintaining integrity the assumptions are made to ensure validity in the results. Below is a list of the assumptions:

1. Constant volume of air within the GH
2. Air flow within the GH is good and the air density distribution is uniform
3. The HP operates at maximum efficiency throughout usage
4. Materials are in a steady state with no internal HSos and have uniform properties across the component

5.2.6 Heat Transfer Model

Building on the foundational setup of the GH, calculations were conducted to assess the system's heat transfer, considering the following external factors; sunlight and differences between external temperatures and the internal temperature. Ground temperatures were measured from 4 inches below the surface. The heat transfer through the wall/roof and ground is calculated initially. This process is carried out by employing Fourier's law, Section 2.2.4, embodied by the equations below:

$$Q_W = \frac{-k_W * A_W * T_i - T_A}{d_W} \quad (13)$$

$$Q_G = \frac{-k_G * A_G * T_i - T_G}{d_G} \quad (14)$$

From here the solar heat gain is determined. This involves inputting the daily kWh from Chelmsford into the equation:

$$Q_S = E_S * \frac{1000}{24} * t_w \quad (15)$$

The overall heat transfer system is illustrated below through a cross-sectional representation of the adjoining wall and ground. The roof shares identical properties with the wall so it is depicted by Q_{wall} .

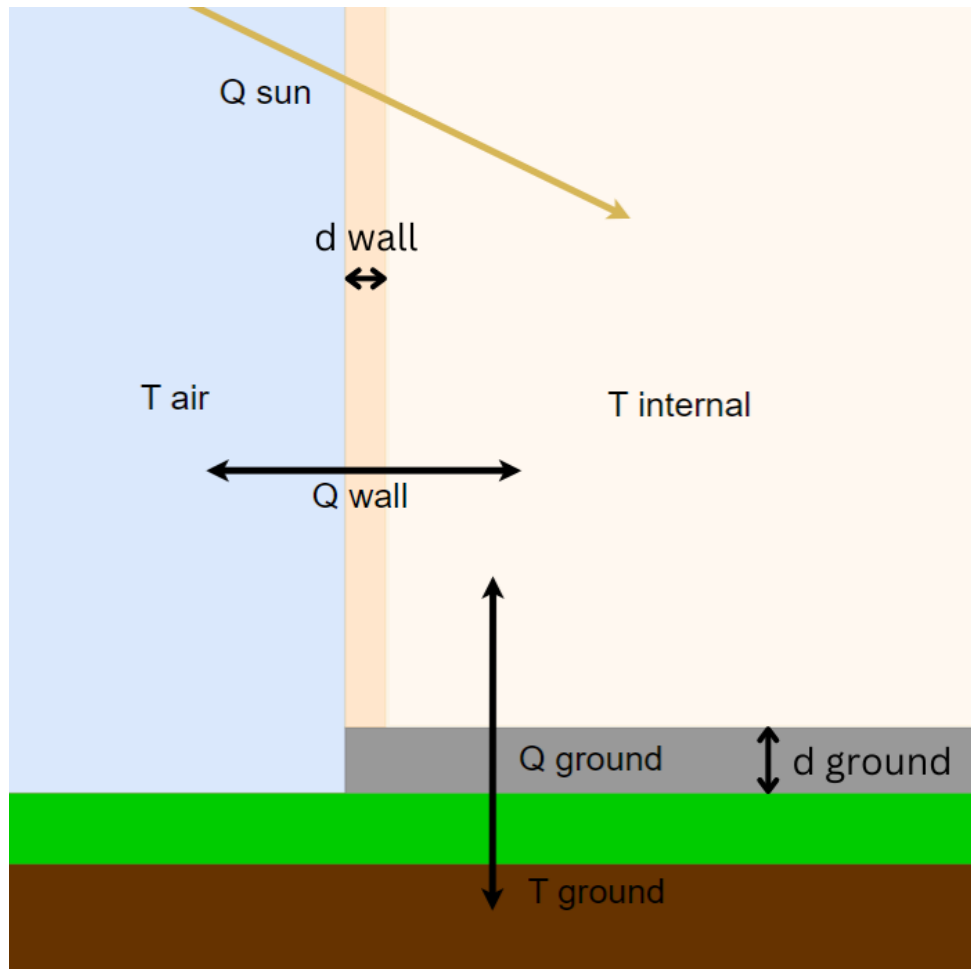
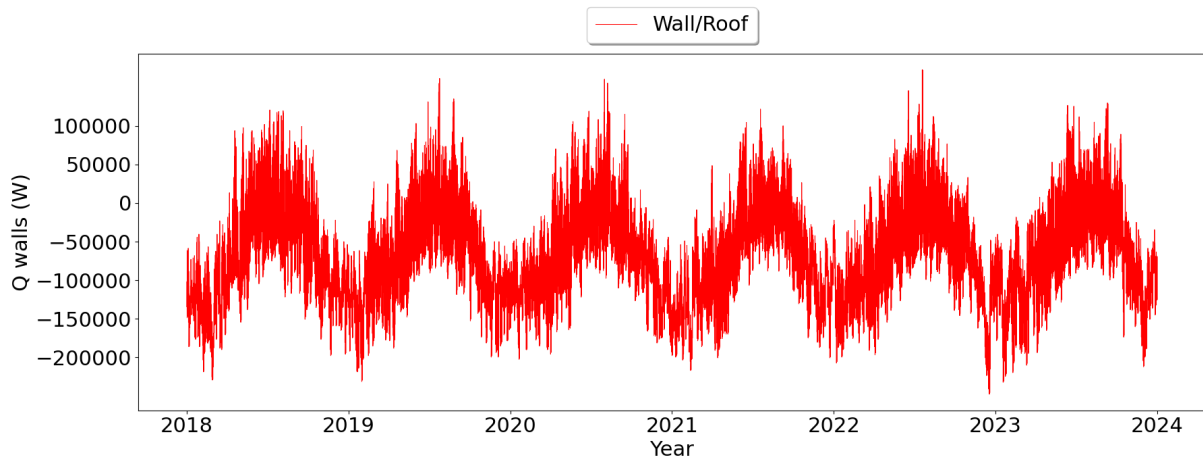
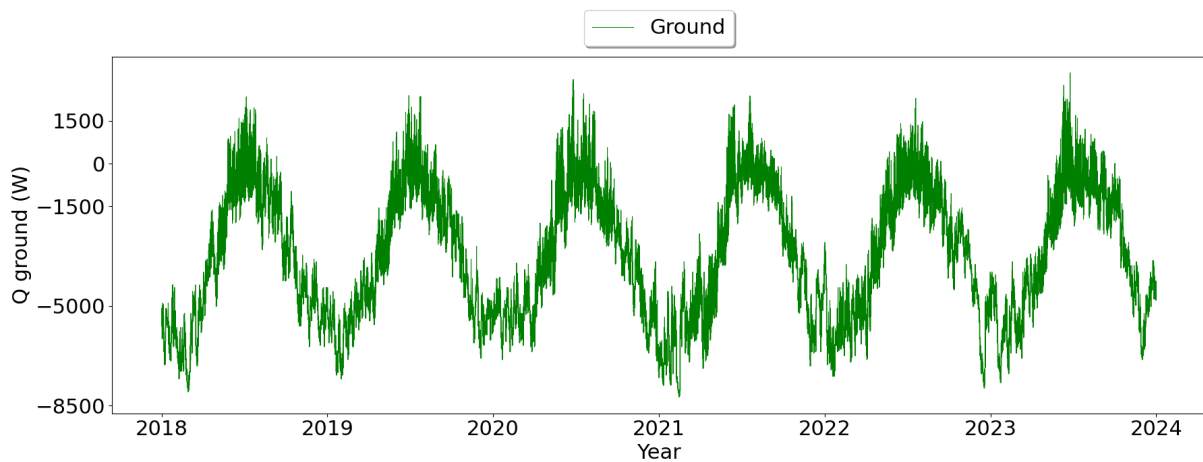


Figure 10: Heat transfer visualisation

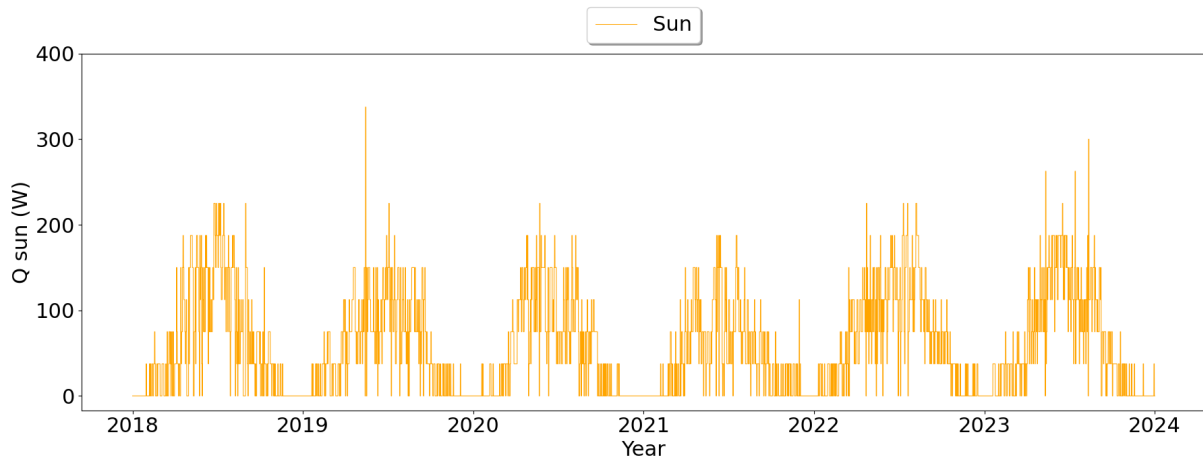
Starting with the wall(shaded dark orange), its thickness is indicated by ' d '. The wall is characterised by a thermal conduction coefficient ' k '. Heat transfer occurs when a temperature gradient exists between the internal and external air temperature, as depicted by Equation 13 and discussed in Subsection 2.2.2. Transitioning to the ground, an identical process applies, represented in Equation 14. Here, ' d ' represents the thickness, which differs from that of the wall, and the ground is also characterised by a distinct thermal conduction coefficient ' k '. The double-sided arrows across the wall and ground are indicative of the reciprocal nature of Fourier's law, see Subsection 2.2.4: if the external temperature (ground or air) exceeds that of the internal temperature then heat will enter the system; conversely, if the internal air temperature exceeds either then heat transfer will ensue away from the system, either through the wall to the air or through the floor to the ground. Lastly, the yellow arrow signifies heat addition from the sun. Notably, this arrow is single-sided, due to the nature of solar heat transfer being non-reciprocal. The transparency of the walls is 0.9 indicating that not all of the sun's energy is able to penetrate the GH. Below is the output of the systems heat transfer derived by employing the mathematical processes described above, expressed by Equations 13 & 14 & 15



(a) Heat transfer through the walls and roof



(b) Heat transfer through the flooring



(c) Heat addition from the sun

Figure 11: Heat transfer across the system

Utilising the external conditions generated in Subsection 5.2.4 e.g. Air temperature ground temperature and sunlight. Over the six year period, the graph portrays that the majority of heat transfer occurs through the walls and roof. This outcome is anticipated, however, as these surfaces encompass the largest surface area exposed to external conditions while possessing relatively thinner widths compared to the ground as they must facilitate sunlight penetration.

The system total heat transfer was calculated by summing the previously described calculations with the output shown in Figure 12:

$$Q_T = Q_W + Q_G + Q_s \quad (16)$$

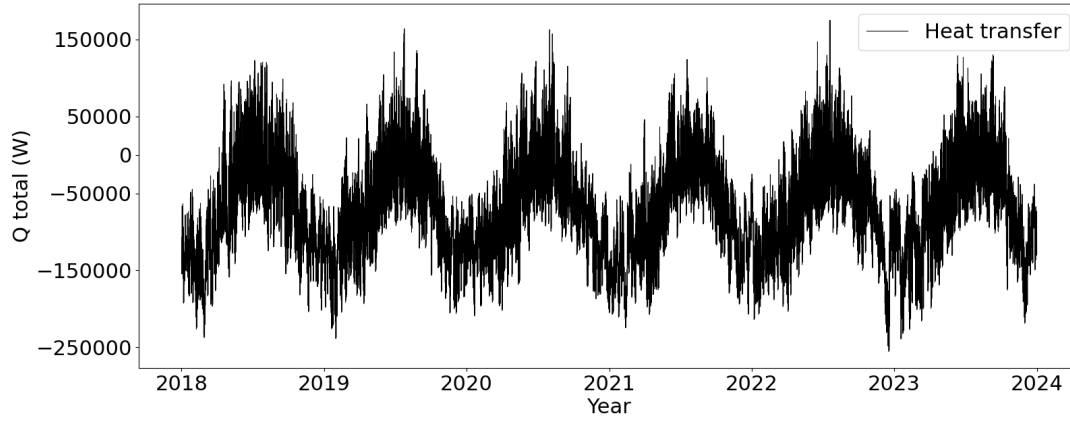


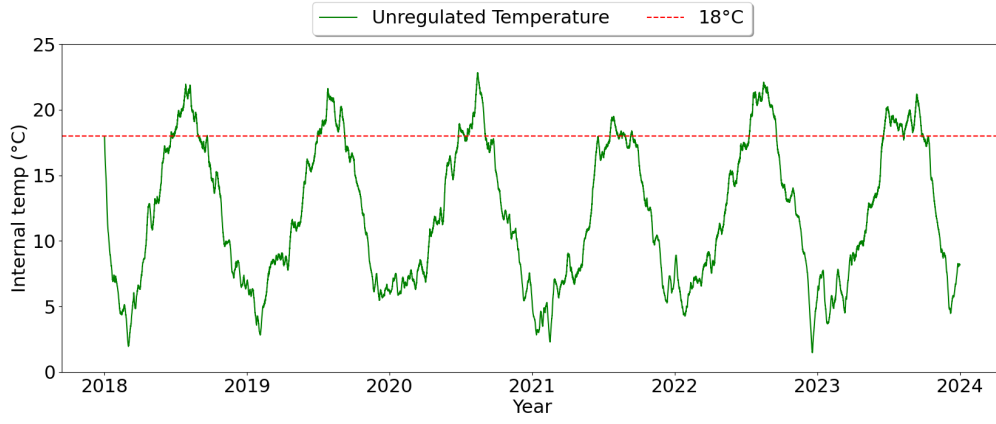
Figure 12: Systems total heat transfer

During the winter months, a significant heat loss is evident, necessitating substantial heat addition by the heat pump to sustain an optimum internal temperature for plant growth. The graph and its corresponding data play a vital role in ensuring the simulation's reliability. From this, the determination of the unregulated internal temperature variation in the absence of heat regulation is calculated by Equation 17.

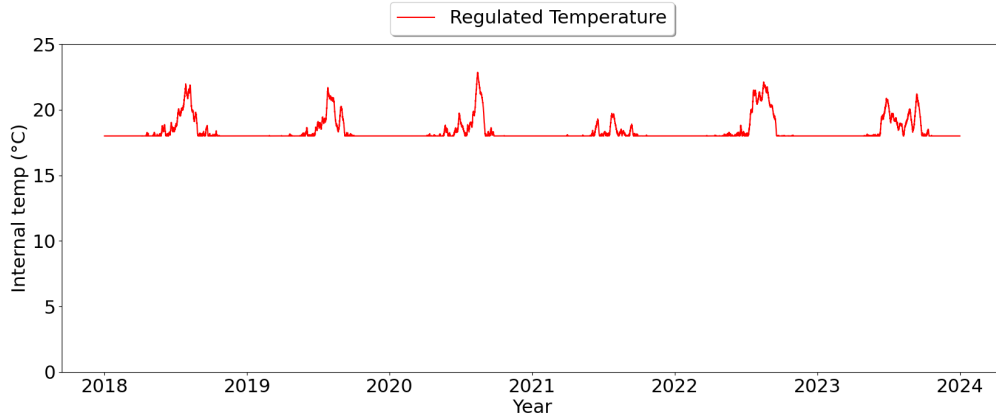
$$T_{Ni} = T_i + \frac{Q_T}{m_a * c_{p,a}} \quad (17)$$

$$T_{Ri} = T_{Ni} + \frac{Q_c}{m_a * c_{p,a}} \quad (18)$$

The outputs of these equations are represented in Figure 13. With the unregulated internal air temperature from Equation 17 in Subfigure 13a and the regulated air temperature from Equation 18 represented in Subfigure 13b



(a) Unregulated Internal Air Temperature



(b) Regulated Internal Air Temperature

Figure 13: Internal Air Temperature Profile

The red dashed line at 18°C represents the target temperature for avocado growth. Throughout the autumn, winter and spring, the GH fails to consistently maintain this desired temperature, suggesting substantial heat addition will be required. This unregulated internal temperature profile illustrates the scenario in which no heat regulation is applied. Conversely, the lower graph displays the regulated internal temperature, as a result of the heat pump compensating for the losses to the surroundings. From here the required heat addition to maintain a lower graph's temperature profile can be calculated. This is done through Equation 19.

$$Q_c = (T_{di} - T_i) * m_a * c_{p,a} \quad (19)$$

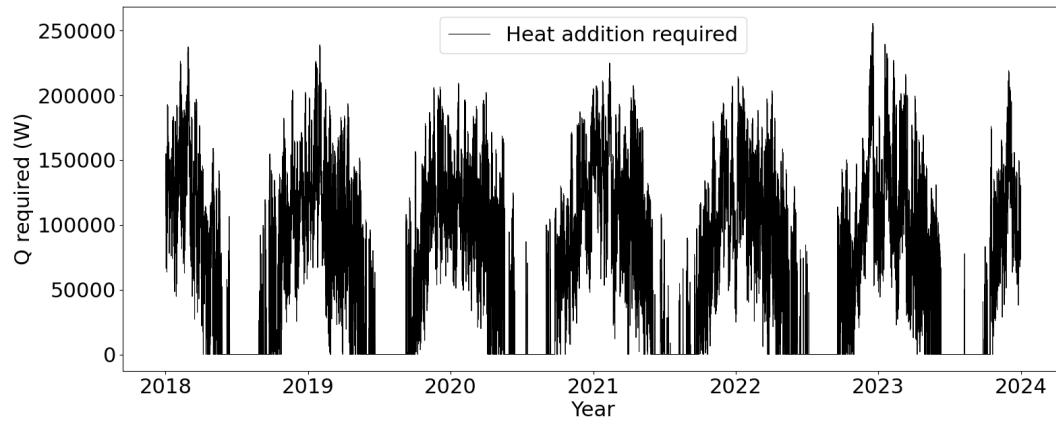


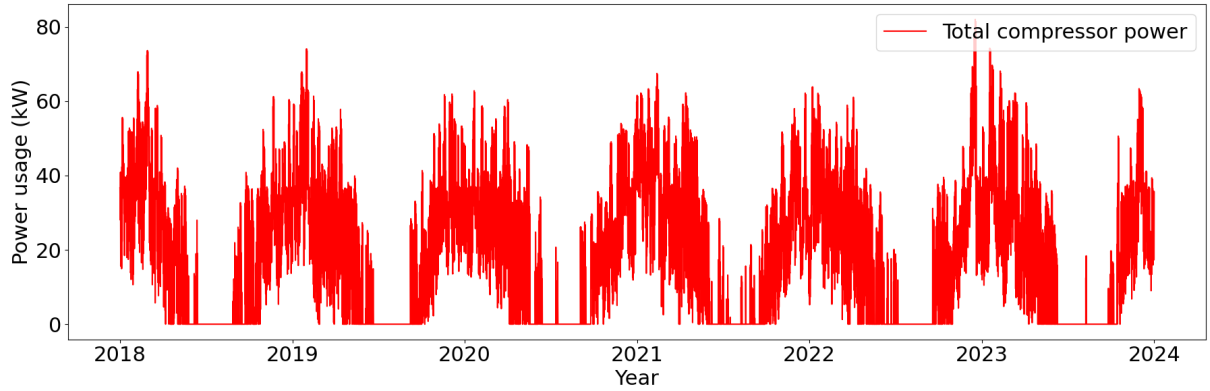
Figure 14: Required heat addition from the heat pump to maintain the internal temperature shown in Figure 13b.

The graph shows the required heat addition from the heat pump to maintain the desired temperature calculated through Equation 19. It is expected that the required heat addition is similar to that of the total heat losses.

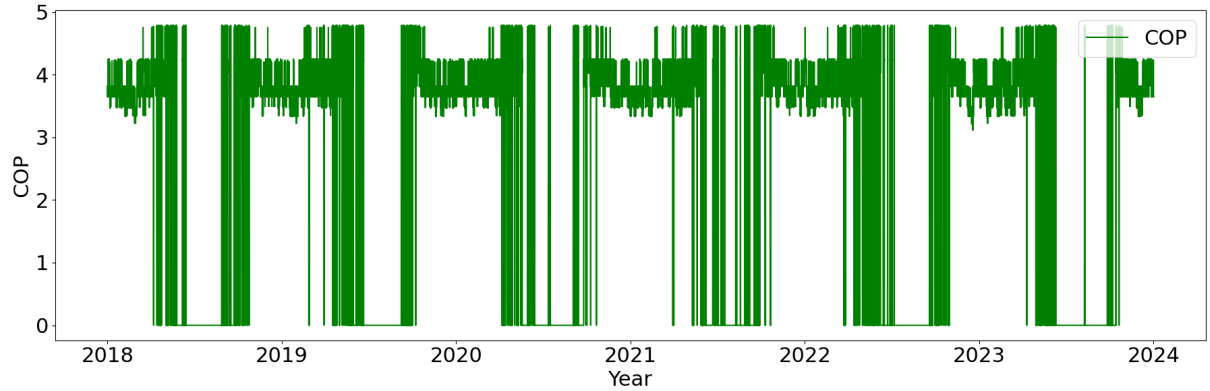
6 Results

The following section presents the performance of the three distinct HP systems (ASHP, GSHP and WSHP), within the context of maintaining optimal temperature conditions for an industrial-scale GH in Chelmsford, UK. Over a six-year simulation period, the operational efficiency was gauged, focusing on energy consumption and system efficiency (Coefficient of Performance-COP). The ASHP, GSHP and WSHP operational period was determined to be 34,400 hours, equating to 65% of the total duration of the simulation period. This operational gauge was used to establish the performance parameters of each HP system.

6.1 ASHP



(a) Total compressors power profile



(b) COP profile

Max COP	Min COP	COP at max Q_c	Max compressor kW
4.79	3.12	3.11	82.03

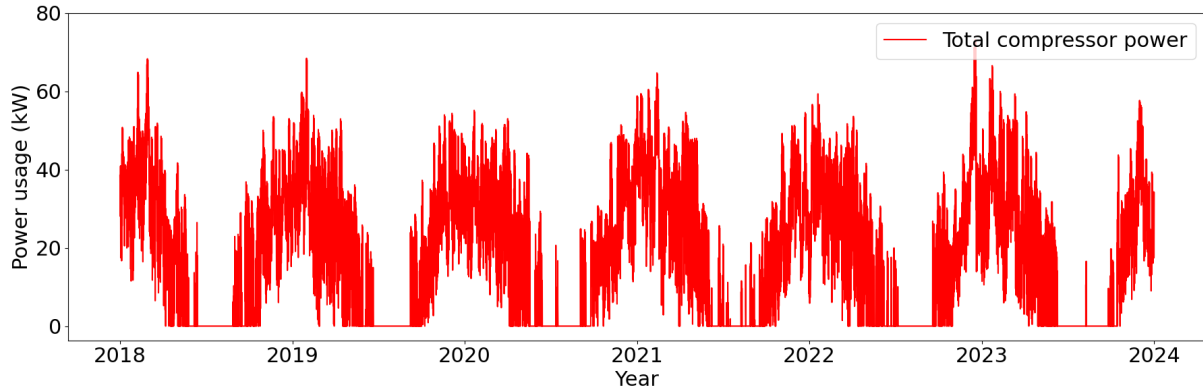
(c) Heat pump performance values

Figure 15: ASHP performance 2018-2024

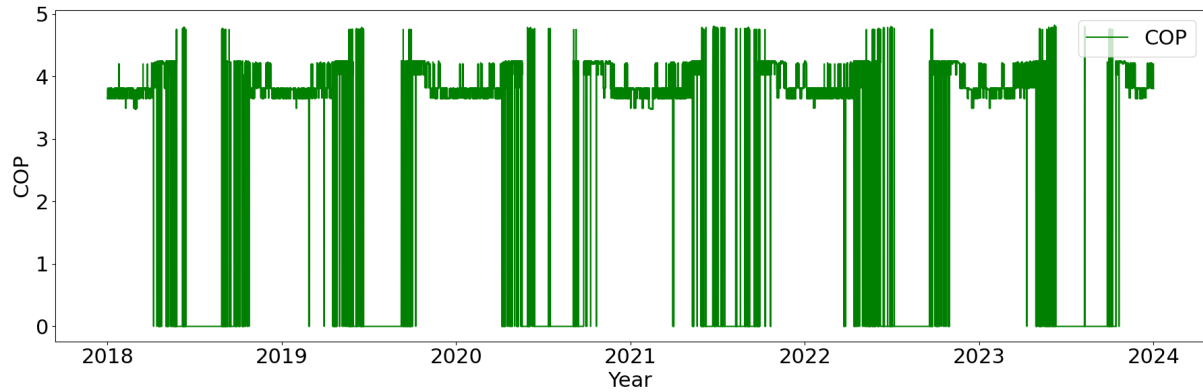
The data shown in Subfigure 15a represents the total compressor power input at each hourly iteration for the ASHP across compressors one and two represented by W_{c1} & W_{c2} in Figure 2. A maximum combined compressor power usage of 82.03kW was recorded, this would be the minimum power rating of the ASHP required to maintain the desired internal temperature.

Subfigure 15b depicts the COP, calculated using Equation 10 over the same period, with an average rating of 4.06.

6.2 GSHP



(a) Total compressors power profile



(b) COP profile

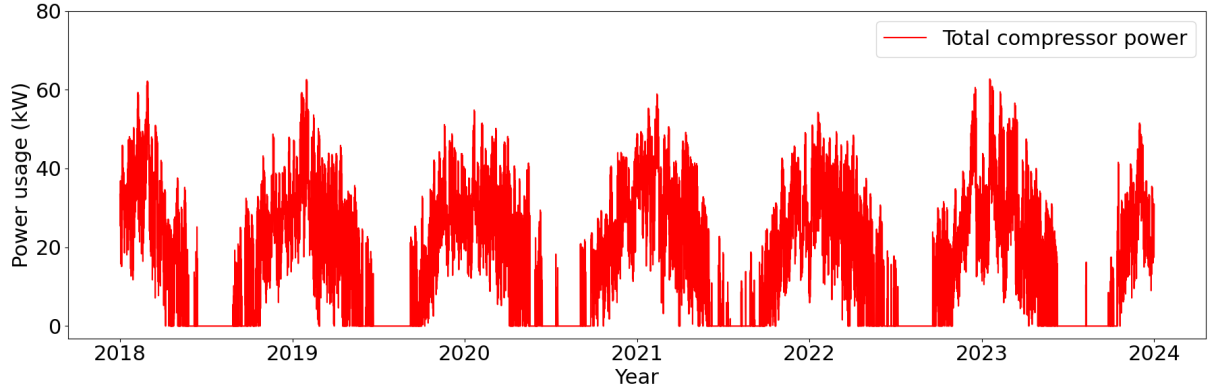
Max COP	Min COP	COP at max Q_c	Max compressor kW
4.82	3.48	3.49	73.72

(c) Heat pump performance values

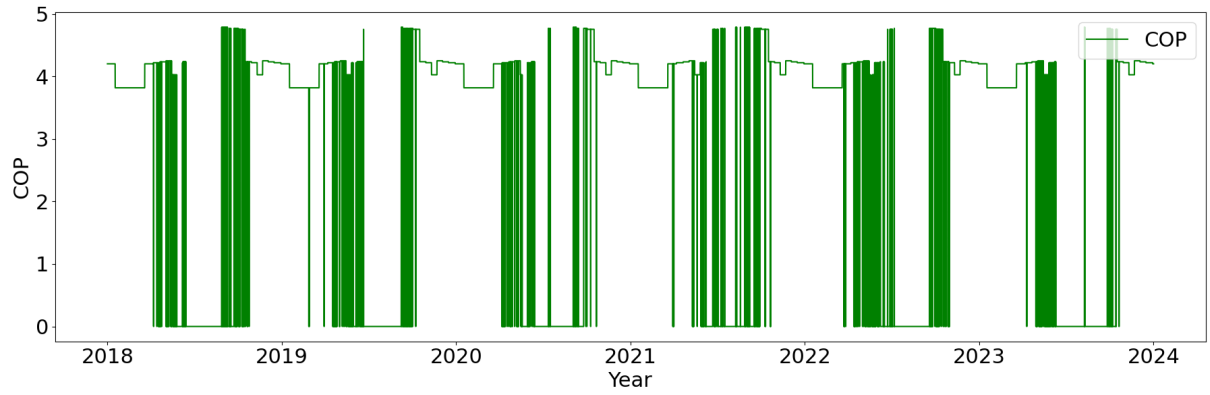
Figure 16: GSHP performance 2018-2024

The data shown in Subfigure 16a represents the total compressor power input at each hourly iteration for the GSHP across compressors one and two represented by W_{c1} & W_{c2} in Figure 2. A maximum combined compressor power usage of $73.72kW$ was recorded, this would be the minimum power rating of the GSHP required to maintain the desired internal temperature. Subfigure 16b depicts the COP, calculated using Equation 10 over the same period, with an average COP of 4.11.

6.3 WSHP



(a) Total compressors power profile



(b) COP profile

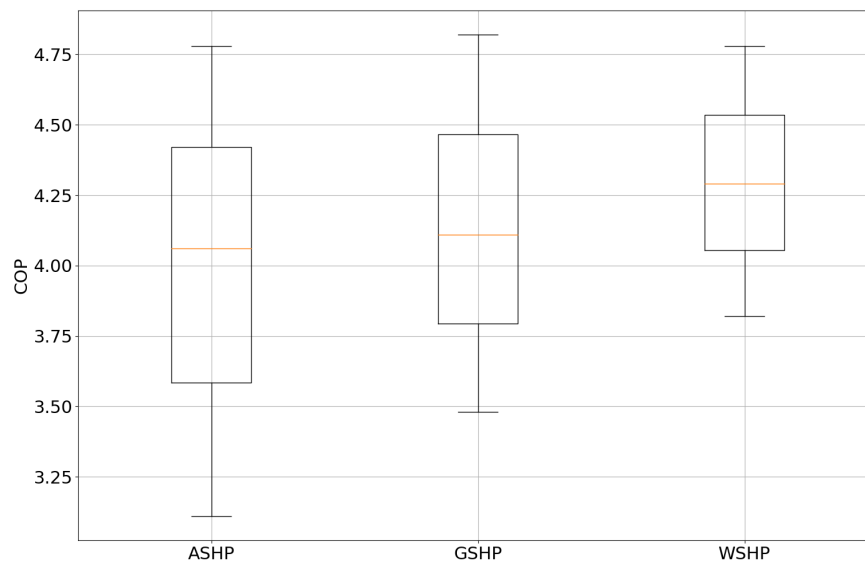
Max COP	Min COP	COP at max Q_c	Max compressor kW
4.78	3.82	4.22	62.71

(c) Heat pump performance values

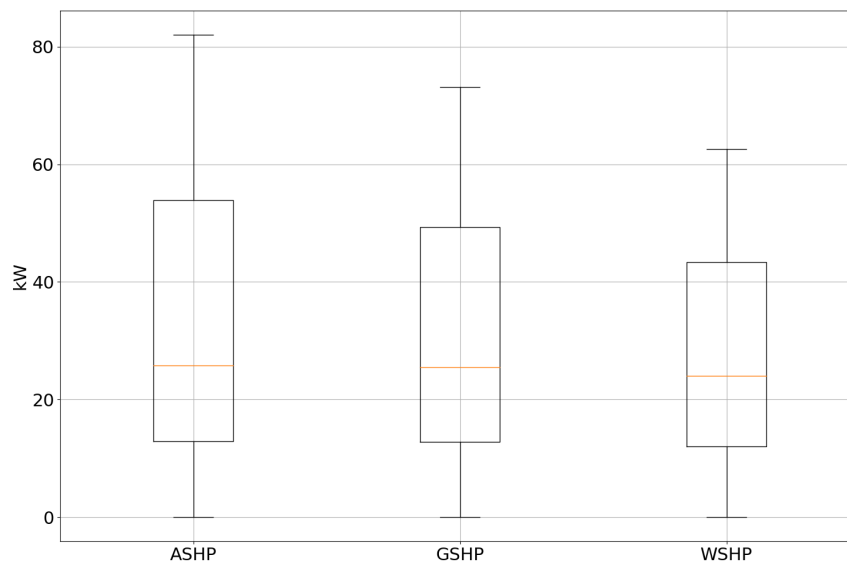
Figure 17: WSHP performance 2018-2024

The data shown in Subfigure 17a represents the total compressor power input at each hourly iteration for the WSHP across compressors one and two represented by W_{c1} & W_{c2} in Figure 2. A maximum combined compressor power usage of $62.71kW$ was recorded, this would be the minimum power rating of the WSHP required to maintain the desired internal temperature. Subfigure 17b depicts the COP, calculated through Equation 10 over the same period, with an average COP of 4.29.

6.4 COP & kW Distribution



(a) COP distribution



(b) Compressor usage distribution

Figure 18: HP performance distribution

The Figures 18a 18b, show the distribution of the COP & kW values for each HP. The median COP for all HPs is above 4, this is a good level of performance. The WSHP has a smaller interquartile range (IQR) than that of ASHP and GSHP indicating it has less variability in its performance. The trends discussed in Figure 18a can be followed in Figure 18b. The WSHP has the lowest median kW rating, indicating the lowest power usage. While the ASHP shows

the largest IQR indicating a more significant variation in power usage. Overall in both Figures 18a 18b there are no extreme outliers indicating all outputs are within an expected operational range.

7 Discussion

The following discussion will discuss several key points following the results.

1. Energy usage
2. COP
3. Heating element
4. Lighting demands
5. Power supply for sustainability
6. Comparison to imports

7.1 Energy usage

To begin the discussion The compressor power will be analysed.

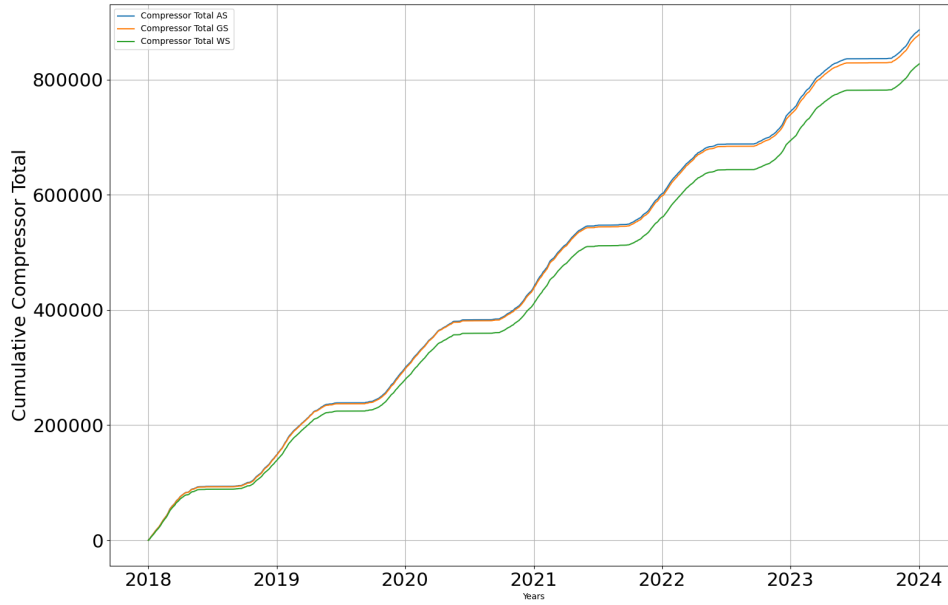
Heat pump	Mean W_{c1} usage(kW)	Mean W_{c2} usage(kW)	kWh/year	Total kWh
ASHP	11.95	13.81	147,675.95	886055.09
GSHP	11.89	13.63	146,320.50	877923.00
WSHP	11.14	12.90	137,843.39	827060.35

Table 7: Energy Usage

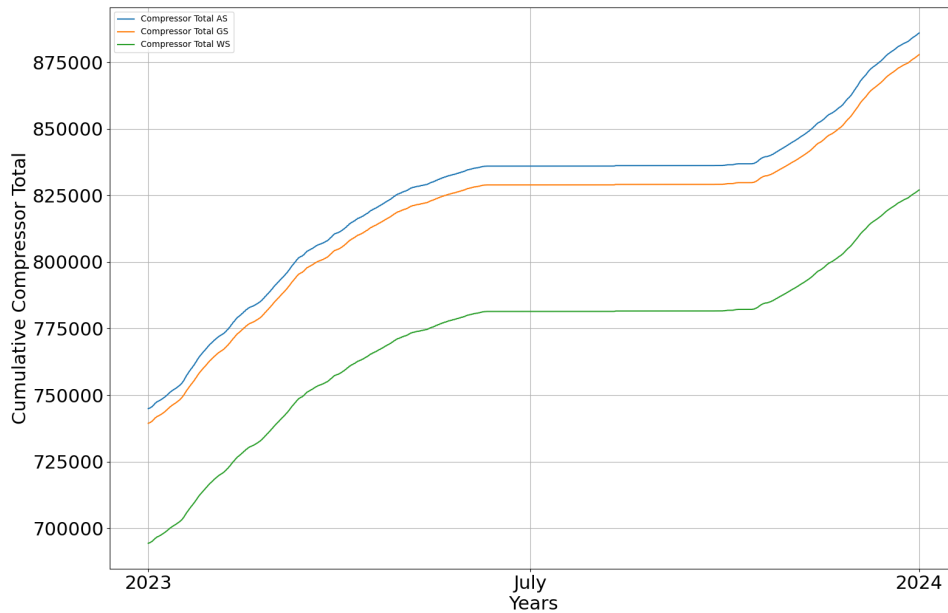
When looking at Figures 15a,16a,17a, a gradual decline from ASHP to WSHP power usage by the compressors can be seen, going from highs of $80 + kW$ to highs of $60kW$. These differences are due to the nature of a HP, As stated in section 2, the larger temperature difference between the HSo and HSi results in a larger amount of work to be done on the system to counteract the natural flow of heat inline with the second law of thermodynamics, Section 2.2.2. The compressor will do the highest amount of work when there is the largest amount of heat addition required while also the highest temperature difference between the HSi and the HSo. For the ASHP and GSHP, this falls on the day that the most heating is required. This is because this is when the most heat is lost due to the natural surroundings, where the HSo temperatures are also the temperatures that contribute to the heat loss of the GH system. A low HSo temperature for ASHP and GSHP means a large amount of heat loss to the surroundings and a large difference between HSo and HSi.

As seen in Figure 9 however, the water source temperatures reach their low in March, therefore, in December when the maximum power input for ASHP and GSHP occurs the water temperature is still relatively warm at $10^{\circ}C$. This means that when a large amount of heat addition is required the WSHP does not require as much work as the others. This is seen by the WSHP doing 27% less work than that of the ASHP and GSHP on the day of maximum heating.

7.1.1 Cumulative Energy Usage



(a) Cumulative compressor usage the six year period



(b) cumulative compressor usage in the final year of simulation

Figure 19: Cumulative compressor usage

By employing the WSHP the energy savings are $60MWh$ from Table 7 and graphically represented in Figure 19 the total economic savings are £13,800 over the 6 years, assuming an average price of £0.23/kWh over the six years [35].

7.2 COP

The heating trends discussed in Section 7.1 can be followed across the main heating months of September to May. As seen in Table ?? the COP of the WSHP maintains a high value while the ASHP and GSHP drop to relatively low values in the main heating months.

Month	ASHP	GSHP	WSHP
January	3.79	3.77	4.02
February	3.86	3.75	3.82
March	3.93	3.82	3.96
April	4.04	4.04	4.22
May	4.16	4.18	4.18
June	4.30	4.42	4.31
July	4.25	4.55	4.63
August	4.28	4.54	4.67
September	4.24	4.23	4.77
October	4.18	4.17	4.47
November	3.96	4.00	4.16
December	3.84	3.80	4.23
Overall COP	4.06	4.11	4.29

Table 8: Monthly COP average

What is notable however is that the WSHP in May and June has either the same COP or lower, than that of the ASHP and GSHP. When looking specifically at the average kW usage over May and June it can be seen that the WSHP uses $34kW$ more in that period than the GSHP. This is a very fine amount. When comparing the usage across the main months of heating of December and January, the compressor power usage is much higher, the performance of the WSHP results in an energy saving of $4000kW$ when compared to ASHP and GSHP.

7.3 Heat Pump Conclusion

From the power usage and COP results, Sections [7.1, 7.2] it is clear to see that the performance of the WSHP is much greater than that of the ASHP and GSHP. The average COP is 4.29 which throughout the 6 years results in a $60MWh$ saving. The GSHP results in a $10MWh$ saving over the 6 years compared to the ASHP, this is not as notable a saving.

7.4 Method of Room Heating

The heating requirements for the room at the maximum are 232,000 Watts. Calculate new water temperature:

$$Q = \delta T * c_{p,w} * m_w \quad (20)$$

Through rearranging Equation 20 to find δT . The water temperature from the inlet to the outlet of the radiator can drop by 61.45°C . From this, the radiator surface temperature can be calculated with the following assumptions So radiator surface temp with the following assumptions:

1. Uniform heat transfer across the flat plate of the radiator
2. Steady-state conditions
3. Ideal conditions
4. Radiator water initial temperature is 35°C

becomes:

$$T_s = \frac{T_{inlet} + T_{outlet}}{2} \quad (21)$$

The surface temperature is 65°C The required heat transfer convective coefficient can be calculated using Equation 3. The result is a convection coefficient of $h = 91.4 \text{ W}/\text{m}^2\text{K}$ when there are 3 radiators of area 18m^2 . Under the laws of convection, this would be considered forced.

Free $h < 5$

Forced $h > 5$

Therefore, The use of fans to circulate the internal air is required to ensure correct ventilation and heating.

7.5 Lighting

For optimal avocado growth, a light intensity of 500 PPFD equating to 6 hours of direct sunlight, is required [36]. However, in Chelmsford during the period November to January, these conditions are not met. Converting PPFD to LUX yields 21739 [37]. Utilising the conversion factor for LUX to solar irradiance, in natural sunlight is $0.0079\text{W}/\text{m}^2$ resulting in required supplementation of $171.74\text{W}/\text{m}^2$ [38]. Given the cultivation area is 400m^2 the power requirements for artificial lighting are calculated to be 68kW , consequently, replicating sufficient sunlight for these months results in a total energy consumption of $37,536\text{kWh}$.

7.6 Power Supply

Taking the ASHP energy usage has the highest energy consumption, the maximum power output is 82kW along with the additional lighting power requirements being 68kW , and the total power requirement at the maximum usage is 150kW . As previously discussed, minimising the environmental impact of the integrated system is essential, thus the chosen power source is wind. The average wind speed is $3.9\text{m}/\text{s}$, v_{wind} , and using the equation for annual estimated production Equation 22 where k is the conversion factor, and r is the radius of the wind turbine swept area.

$$AEP = k * v_{wind} * \pi * r^2 \quad (22)$$

When combining the annual energy usage of the lighting and heating requirements the minimum AEP is $185,211.95\text{kWh}$. It was found that the required radius for the wind turbine to achieve these power requirements would be 17.62m . However, this assumes that the turbine is operating at maximum efficiency. But in reality, there is a theoretical maximum for wind turbines known as the Betz limit [39]. The Betz limit is the theoretical maximum efficiency for a wind turbine's

ability to convert wind energy into mechanical energy. It states that no wind turbine can capture more than 59.3% of the kinetic energy in wind. Dividing the calculated radius by the Betz limit, the new radius is 29.86m. The maximum usage at a given time is 150kW, Upon researching commercial wind turbines it was found that 150kW rated wind turbines are 25-32m in radius, in line with the calculation presented here. The Wind World W2700 turbine has been identified as suitable for the area, with a cut-in speed of 2.0m/s and a rated power of 150kW. These turbines are very commercially available both new and second-hand. Secondhand would limit the environmental impacts further while settling for hindered efficiency, you could purchase 3 of these turbines for a combined £30,000, This would be recouped in energy savings within a year, while also being able to power a minimum of 2 GHs year-round. This means the cost of running the GH after 6 years of usage would be minimal with energy production taken care of.

This study recognises the inconsistency of wind power, and that the alignment of maximum wind energy production will not fall on the days when the maximum energy usage from the GH does. However, innovative storage techniques in the field of thermal storage integrated with wind power would help to negate these inconsistencies and ensure a constant supply of renewable electricity. A study by Neumeier et al. [40] explores the idea of converting wind power directly into a stable heat energy source through wind thermal turbines. However, these novel ideas are not yet commercially scaled. The novel idea however poses a promising future for wind energy and the feasibility of the GH. With research still required, the heat energy from the wind turbine could be stored for later conversion to electricity for the HP or bypass the HP system directly into the heating element of the GH. Either way, both approaches provide promise for the long-term sustainability of the integrated GH system.

7.7 Avocado Comparison

Considering the article discussed in Section 3.3 of the literature review, the carbon footprint of a tonne of avocados was analysed. It was found that the growing and shipment of a tonne of avocados from South Africa to Nottingham produced 904kg/CO₂. The GH presented here has the facilities for 9 avocado trees when correct spacing is taken into account as each tree requires a 3m radius, each producing 100kg/year [41]. The total production of avocados would be 900kg. Without a thorough LCA of the proposed integrated GH system, it is hard to determine the overall emissions. However, the study by Blauww et al, considers farm electricity, machinery and transport to be the prime contributors to the total emissions per tonne of avocados. Analysing the integrated system proposed in this project, the transport emissions are negligible as no shipping is required bar electrified rail freight. The GH electricity demands are met through renewable sources (wind). The overall emissions of the GH in this case are assumed to be 0. Thus it was calculated with the production rates and assuming both the South African farm and the Chelmsford GH use the same fertilizer and pesticides, the integrated GH system if employed would save 630kg/CO_{2e}/tonne of avocados. This is a preliminary estimate as factoring in the manufacturing, transport and fitting of the integrated system will have an environmental impact.

8 Wider Context

8.1 Avocado Preliminary Emissions Savings

On average each person in the UK consumes 1.6kg of avocados per year. With the population at 67 million people, this would result in 107 million kg of avocados consumed annually. With each GH able to produce 900kg, it would require 119,000 GHs or a 47km² area covered to produce the entire UK's avocado demand. This area is equivalent to roughly a quarter of the size of greater Glasgow. Furthermore, the estimated emission savings stated in Section 7.7 applied to this scenario would result in 64.5 million kg of CO₂ saved from entering our atmosphere.

A report by The Guardian indicates that a round-trip flight from London to New York generates approximately 986 kg of CO₂ per passenger [42]. By implementing this GH system exclusively for avocado production to satisfy the UK's demand, CO₂ savings could effectively offset the emissions of more than 65,000 passengers travelling this route. This demonstrates the significant environmental benefits of adopting the proposed GH system within the UK.

No LCA has been conducted therefore the environmental implications of the system are estimated. However, the impacts analysed in this discussion while preliminary are promising as the UK would be able to increase its food security using a relatively small percentage of its land and large emissions savings. The usage of this GH is not restricted to avocado growth, the large dimensions ensure that the growth of many other fresh produce could be produced year-round.

8.2 Other Potential Production

Currently the UK imports 111.78 million kilos of lettuce from Spain the 32nd most water-stressed country on the globe [43]. Lettuce itself is 96% water, it seems counterintuitive to import 107,308,800kg of water from Spain when there is a vast amount in the UK's isles. A study by Barbosa et al. [44] concluded that the hydroponic GH growth of lettuce resulted in a production rate of 41kg/m²/year. Applying this growth rate to the integrated GH system discussed in this simulation would require a 2.72km² area to produce the equivalent of the UK's yearly lettuce import from Spain. This would improve the food security of the UK while reducing the stress of the Spanish water infrastructure. While further analysis would have to be carried out to determine the energy requirements for the different growing conditions of lettuce to that of avocados it indicates promising results.

9 Conclusion

Analysis of the heat transfer system, HP performance and power requirements indicates the feasibility of this sustainable GH model is positive. Its impact on the food security of the UK and the global environment would be of significance should real-world implementation, scaling up, and the factors in Section 9.1 be addressed. The WSHP performed best in maintaining the optimal growing temperature for Avocados. This was due to the stability of the river water's temperature in comparison to that of the ground and air. The GSHP performed better than the ASHP. Before employing any of these technologies financial costs, environmental impact, CO₂ emissions and wider impacts relating to manufacturing, installation and maintenance would all need to be taken into account to determine the viability and the most appropriate HP-type location-specific. The following points are a summary of the project work and its outcomes:

1. WSHP has the highest COP and lowest kWh usage over the six years, saving £13,800 compared to ASHP and GSHP in this model
2. Maximum power demands of 82.03kW for the ASHP, 73.72kW for the GSHP and 62.71kW for the WSHP
3. These values result in the following annual heating demands: ASHP = 147,000kWh, GSHP = 146,000kWh, WSHP = 137,000kWh
4. 68kW of additional lighting required throughout November to February
5. Up to 150kW of electricity required for this industrial GH in the UK climate
6. 150kW wind turbine required for annual energy usage
7. Coverage of 47km² to produce UK avocado demand
8. Food security would be improved but a large area is needed, and wider impacts of vast land usage need to be considered.
9. Potential usage for other fresh produce the UK imports on mass, further research needed

The results for the HP are in line with those discussed in Section 3.5. It is evident that as a nation the UK must act in a timely fashion to prevent its food supply chain from collapsing. The sustainable integrated system proposed in this report provides a theoretical groundwork for modernising aspects of the UK's agricultural industry while offering the prospect of significantly improved food security.

9.1 Future Work

This project encompasses a large array of real weather data and accurate heat transfer modelling. However, several areas would need further analysis to bring this proposal from an idealised theoretical proposal to a real-world applicable system. A number of these factors are outlined below:

1. Develop a more rigorous heat transfer model, this can be done by implementing complex calculations to reduce the number of assumptions, for example, analysing the behaviour of the PVC panel and the connection points to determine a more accurate heat loss value.

2. Optimisation of the GH parameters, e.g. explore the impact on energy efficiency and growth rates with differing dimensions, designs and materials.
3. Perform computational fluid mechanics to understand the airflow within the GH such as the heating system interface with the airflow, This may reveal inefficiencies that impact performance.
4. Applying efficiencies to the components in the heat pump and accounting for these losses at each stage.
5. Further complexity of temperature variation impacts, for example, the highest and lowest temperatures are not always at midday and midnight as seen in Figure 7.
6. Conduct an LCA on the entire integrated system to gain a better understanding of the environmental, financial and human impacts of the system.
7. Analyse and optimise growth rates of plants under the varying conditions that can be produced by the GH.
8. Construct a techno-economic report of the optimised system to put forward to investors and governments.

By undertaking these tasks the proposed system has the potential to go from a theoretical basis to a real-world solution to bring stability to the UK's food supply chain in a labile and changing environmental and geopolitical landscape.

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11 Appendices

A Jupyter Notebook Simulation Code and Values

<https://www.dropbox.com/scl/fo/8rqsvtjsiwt6lquwa19v5/AGW17VInZ3I12-5mMY33rlkey=bd6298mmrvskncmd9bn34wfq1&dl=0>

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